

Practitioner's Guide to ACL

A comprehensive guide for practitioners leveraging technology to optimize anterior cruciate ligament recovery and return-to-performance planning.



Knee Extension Seated

Max Force Left & Right Side

Left

590 N

Right

549 N



Single Leg Squat

Avg Peak Knee Flexion - Left & Right

Left

92°

Right

83°



Countermovement Jump

Max Eccentric Deceleration Impulse

Left

147.9 N s

Right

120.3 N s



About the Author

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  @paulread1010

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Previously, Paul has held roles as the General Manager of the Institute of Sport, Exercise and Health (London), Clinical Lead Researcher and Head of the Athlete Assessment Unit at Aspetar Sports Medicine Hospital (Qatar) and Program Director of Strength and Conditioning at the University of Gloucestershire (UK).

Paul has a PhD from Cardiff Metropolitan University in injury risk management and a master's degree in Strength and Conditioning from Middlesex University. **Paul is an accredited strength and conditioning coach with both the UKSCA and NSCA and has consulted with Olympians, professional and international athletes in a range of disciplines.**

Some Words from Other ACL Experts



Dr. Roula Kotsifaki

PhD in Biomechanics,
Physiotherapist

Aspetar

   @roula.kotsifaki

“

A criteria-based approach ensures rehabilitation progresses at a pace that aligns with the athlete's capacity to handle the load, optimizing safety and effectiveness.

This data-informed monitoring removes guesswork, providing a clear, personalized roadmap for recovery.”



Mick Hughes

Physiotherapist, ACL
Rehab Expert, Founder
Learn.Physio

   @mickhughes.physio

“

The rise of technology-enabled objective measurement in ACL rehabilitation has made a large impact in improving outcomes and reducing the reinjury rate for those returning to sport.

The objective data from DynaMo and ForceDecks has been a game-changer in my practice, helping me make more informed clinical decisions to confidently progress my ACL patients and athletes through their rehabilitation journey.”



Jane Rooney

Clinical Director,
**Prahran Sports
Medicine Centre**

[in](#) [f](#) [@jane.rooney007](#)

“

With contemporary evidence recommending criteria-based progression following ACL injury, **the use of technology is becoming imperative in assessing the rehabilitation pillars of strength, power and neuromuscular control.**

Tools such as force plates and dynamometry identify critical deficits, such as quadriceps weakness and balance impairments, early in recovery.

While the addition of motion capture technology assesses complex neuromuscular and neurocognitive skills in the later stages of recovery.

These technologies guide testing and rehabilitation progressions through each stage, from restoring movement to building strength and power and neuromuscular control for a safer return-to-sport. ”



Kate Beerworth

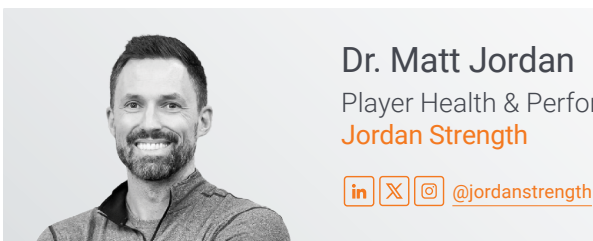
Physiotherapist,
Cricket Australia

[in](#) [X](#) [@KBeerworth](#)

“

VALD technology has **transformed my approach to ACLR rehabilitation by providing objective data and real-time monitoring of key return-to-play metrics.** It allows for personalized, effective rehabilitation, with clear feedback to keep patients engaged.

Tracking trends from jump data and strength helps identify deficits, support long-term knee health, and reduce reinjury risk. Technology has significantly improved my practice, ensuring better outcomes for ACLR patients. ”



Dr. Matt Jordan

Player Health & Performance Consultant,
Jordan Strength

[in](#) [X](#) [@jordanstrength](#)

“

We are all interested in finding the magic 8 ball that answers all our questions about reinjury risk reduction and prevention. But these sorts of questions are very hard to answer. A performance focus, on the other hand, is pretty simple.

We train our athletes to increase their physical and mental capacities so that they can out-sprint, out-jump and out-manoeuvre their opponents while solving motor problems on either limb, no matter the pressure, the stakes or the context.

Instead of thinking about return-to-play testing as your magic 8 ball that ‘predicts reinjury risk,’ think about it as assembling puzzle pieces that help to reconstruct an athlete’s performance profile. ”

Introduction

Anterior cruciate ligament (ACL) injury is a high-profile injury, given that sports with elevated risk are typically the most popular globally. These sports include:

- Football codes (soccer, American Football, Australian Rules Football);
- Basketball;
- Handball;
- Rugby Union;
- Netball; and
- numerous other sports where twisting, pivoting and change of direction (COD) movements are common.



...a 9-month recovery is the minimum time for most athletes to return to sport safely, and this may be extended to 12 months for recreational athletes.

The recovery period from these injuries is long, resulting in a substantial loss of playing or activity time. It is generally suggested that a 9-month recovery is the minimum time for most athletes to return to sport safely, and this may be extended to 12 months for recreational athletes.

Combined with the long recovery time, reinjury to either the ipsilateral or contralateral leg is reported to be around 15%, resulting in an extended period of inactivity.

...reinjury to either the ipsilateral or contralateral leg is reported to be around 15%...

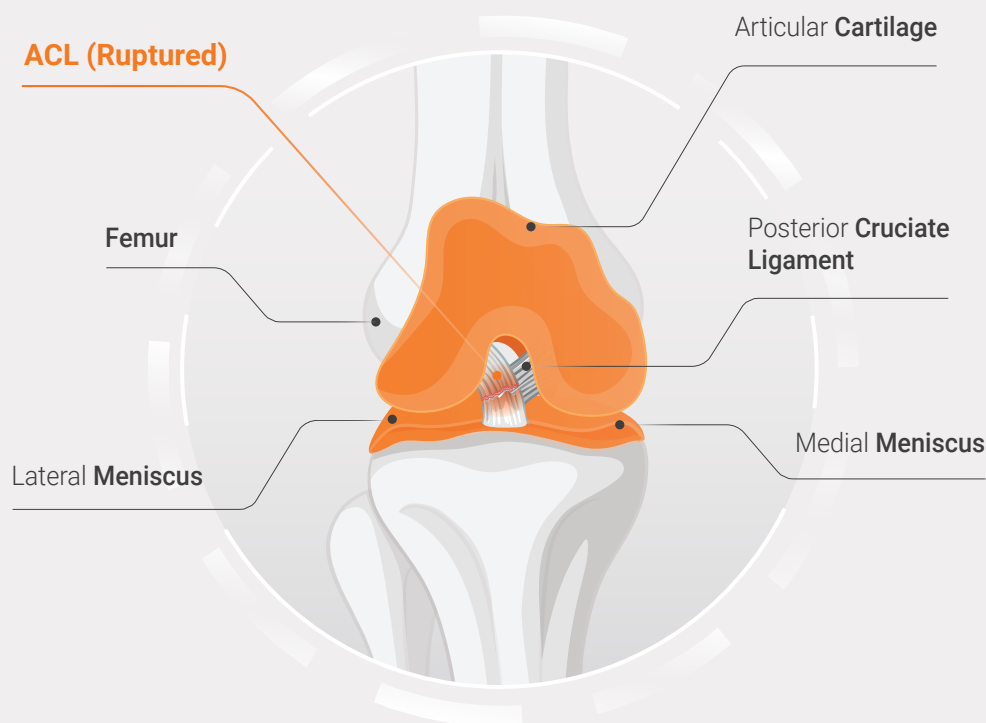
The typical management of ACL injury involves reconstructing the ligament with a view to restoring the stability of the knee. This is typically done through the harvesting of tissue from a donor site within the injured person (autograft). Several options exist, including hamstring, patellar and quadriceps tendons.

An alternative tissue replacement to an autograft is an allograft, where a surgeon harvests a ligament from another person (most commonly a cadaver).

Decisions around graft type selection are multifactorial, with pros and cons weighed for each option. Discussing these is beyond the scope of this guide, but a shared decision-making process is recommended.

In some instances, other structures in the knee are also injured and subsequently surgically repaired. These include the collateral ligaments, meniscus and postero-lateral corner of the knee.

These factors may collectively influence the initial management plan and recovery trajectory. Where applicable, this guide will outline their impact.

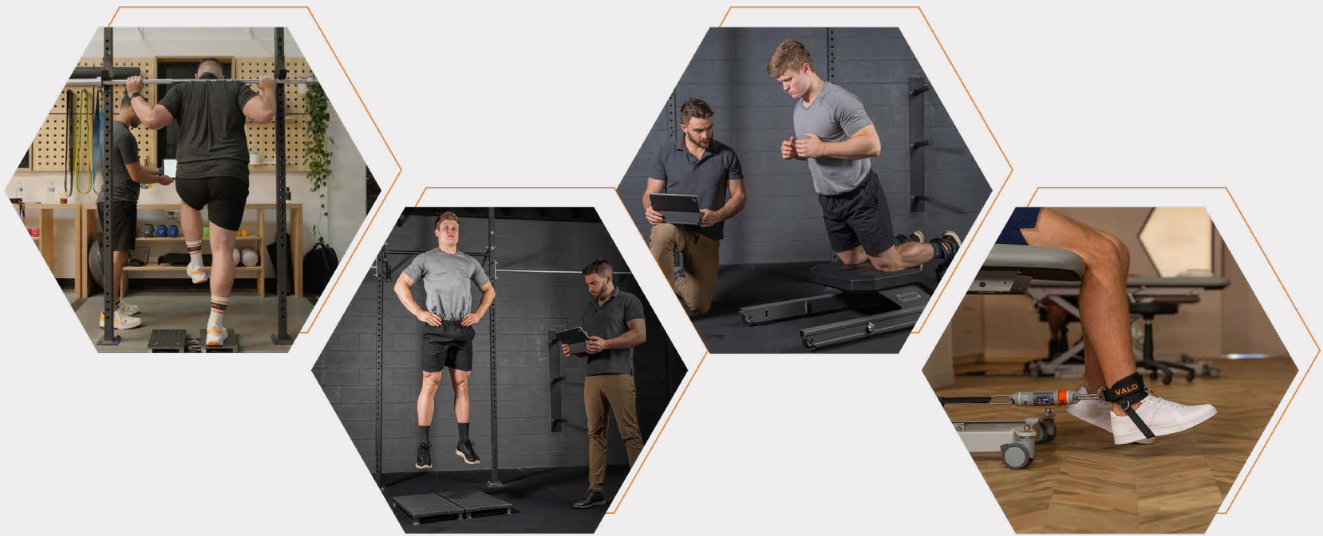


Knee anatomy highlighting key structures and a complete ACL rupture.

...a criteria-based approach [is] used to help guide a person [to safely execute] **key clinical milestones, including running, COD, return to training and ultimately a successful return to competition.**

In recent times, there has been increasing consideration of non-operative management in grade III ACL injuries depending on the requirements of the injured person. Growing evidence indicates the ligament **can heal** given the right conditions. The recovery period for non-operative management is still lengthy and requires an equally thorough rehabilitation.

While biological healing of the tissue is critical, recent work has begun to focus on a criteria-based approach, with a greater focus on restoring physical characteristics. These criteria are used to help guide when a person is capable of safely performing key clinical milestones, including running, COD, return to training and ultimately a successful return to competition.



Similarly, there is a goal to identify factors that contribute to athletes returning to a level of performance similar to pre-injury levels while reducing the risk of reinjury.

At the core of a criteria-based approach is the accurate measurement of physical properties, an essential component to guide practitioners in their decision-making.

Access to measurement technology has been enhanced by user-friendly software, such as [VALD Hub](#) and the VALD objective measurement suite, allowing efficient data capture, simple processing and data interpretation that has been previously confined to university laboratories.

This guide outlines the applications of technology throughout the ACL recovery journey, helping practitioners optimize its use to collect accurate data that informs treatment planning and decision-making.

This guide [helps]...optimize [technology] use to collect **accurate data that informs treatment planning and decision-making.**

The guide is broken up into sections so that you are able to easily incorporate these principles with any athlete who has an ACL injury.

We hope this content provides insights into a modern approach for assessing athletes as they aim to return to performance.

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In this document, we will **cover a wide range of topics and answer some common questions**, including:

Click any heading to jump to section

Guiding **Principles**

Change of Direction Performance

Case Studies

Movement Assessment
in **Early Rehabilitation**

Squat Assessment

Single Leg Squat Assessment

Step Down (Land and Hold)

Key Physical
Characteristics

Jump
Assessments

Understanding
Movement Strategy

Range of Motion

Quadriceps Strength Deficits

**Optimizing Strength
Assessment**

ACL Rehab Timeline and Benchmarks

What Really Matters in ACL Rehabilitation?



Current evidence shows that our understanding of the physical requirements for successful ACL recovery is still evolving.

Given the high reinjury rates, we should question whether current guidelines truly prepare athletes to return to peak performance.

Technology now allows us to track the restoration of key physical attributes, helping to bridge the gap between rehabilitation and performance.

There is a need to...develop strategies that better determine physical function to **“bridge the gap” between rehabilitation and return to performance.**

Accurate and meaningful data collection is essential to guide treatment planning and decision-making, particularly during the extended recovery process of ACL rehabilitation.

Returning to sport should be viewed as a comprehensive journey, not one defined by a single test or specific set of assessments.

This guide emphasizes the importance of clinical reasoning, with testing and technology serving as tools to complement and enhance the decision-making process.



Limitations of Current Testing Protocols

After ACL reconstruction (ACLR), residual deficits in maximal strength, explosive strength (i.e., rate of force development (RFD)) and reactive strength are commonly reported despite having clear rehabilitation guidelines, including criterion-based progressions.

However, the most common method used to measure return to sport readiness has been through various [horizontal hop tests](#).



Expert Tip:

Identifying reinjury risk using outcome measures such as distance hopped lacks sensitivity and cannot characterize persistent mechanical and [movement strategy deficits](#) commonly observed [following ACLR](#). By solely focusing on this, specific physical characteristics that contribute to movement can be overlooked.

This approach also does not provide insight into compensatory movement strategies that may need to be addressed in a complete and thorough rehabilitation.

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An example is provided below. This athlete was tested at five time points following ACLR. Jump height increased linearly at each assessment; however, eccentric braking force characteristics (measured via force at zero velocity) were not targeted effectively during their rehabilitation.

Subsequently, no meaningful changes were observed despite an increase in jump height. Of note, this athlete returned to sport shortly after their last test and re-ruptured their ACL within the first two months.

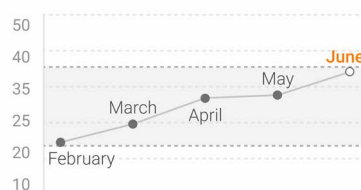


Countermovement Jump

✓ Jump Height — Flight Time (cm)

36.2cm

↑ 11.4%



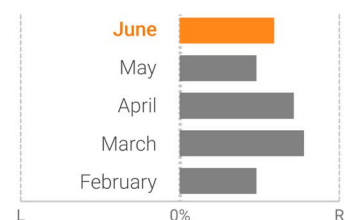
Countermovement Jump

✓ Force at Zero Velocity - Asym

Right
21%

24%

Left 768N
Right 930N



There is a need for more thorough assessments of athletes' functional performance and movement strategies after ACLR, utilizing technology that is both time-efficient and cost-effective.

Frameworks aim to optimize assessment methods and support decision-making for rehabilitation, sports science and strength and conditioning coaches.

They help focus on identifying and evaluating common deficits in physical capacities after ACLR or ACL rehabilitation while highlighting achievable performance metrics to guide data collection and strategies for full recovery.

Basic Framework for **ACL Rehabilitation**



Guiding Principles

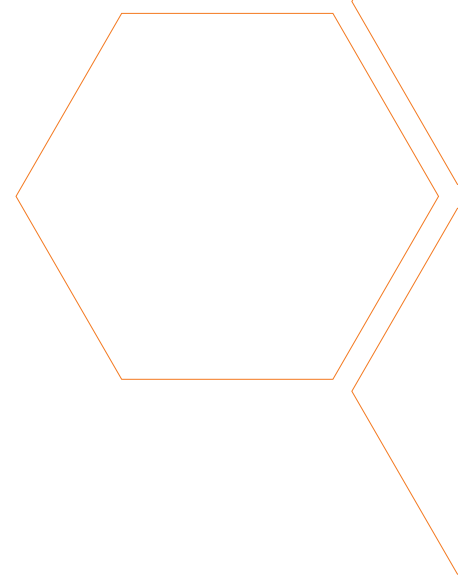
A wide range of physical characteristics are relevant for athletes returning to sport after ACL injury or ACLR. Common residual deficits and subsequent assessment strategies are important to identify and address early in ACL rehabilitation.

Creating longitudinal assessment pathways for monitoring athlete progress is another key element of successful rehabilitation. While the details of ACL rehabilitation appear complex, underpinning values emerge as themes that contribute to successful ACL rehabilitation outcomes.

Practitioner's Guide to ACL Injury Checklist



- ☐ Collect accurate and reliable data.
- ☐ Interpret results to understand the individual's level of physical function.
- ☐ Create a data-informed treatment plan for individuals to return to sport.
- ☐ Monitor and retest to ensure progress is realized.



Advancements in technology have significantly enhanced the precision and scope of assessments following ACL injury. Technology adds value by:



Objectivity

Quantify knee function through objective assessment and training data.



Monitoring

Technology-enabled systems allow for integrated testing within training sessions. Enhancing athlete monitoring capabilities.



Asymmetry

Calculating asymmetries between left and right sides and identifying strength imbalances within the joint.



Criteria

Make better decisions during rehabilitation and return to play scenarios with criteria-based rehabilitation.



Biofeedback

Real-time input into movement strategy helps to correct movement deficits early in rehabilitation.



Benchmarking

Enable greater insights with age-matched normative data for athletes.

Key Physical **Characteristics**

Range of **Motion**

In the pre-operative and early phases of rehabilitation, the primary aims are to minimize swelling, restore range of motion (ROM) and reestablish contraction of key muscles to ensure a “quiet knee.” Objective measurement of knee ROM is an important part of this process.

Achieving full knee extension ROM preoperatively reduces the chance of postoperative complications. Recovering knee extension ROM early is essential to normalize gait in the first few weeks following surgery and reduce the risk of developing anterior knee pain, often linked with poorer long-term outcomes.

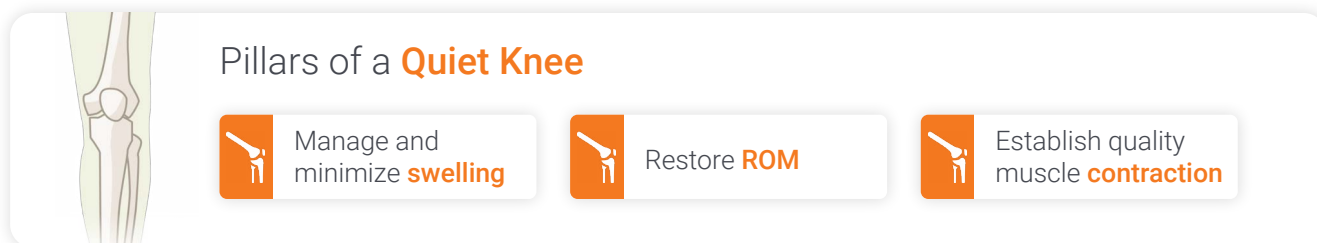


Recovering knee extension ROM early is essential to normalize gait...and reduce the risk of anterior knee pain [leading to] poorer long-term outcomes.

When measuring ROM with systems such as [DynaMo](#), it is recommended that full active knee extension is achieved before progressing to heavy strength, neuromuscular training and running-based activities.

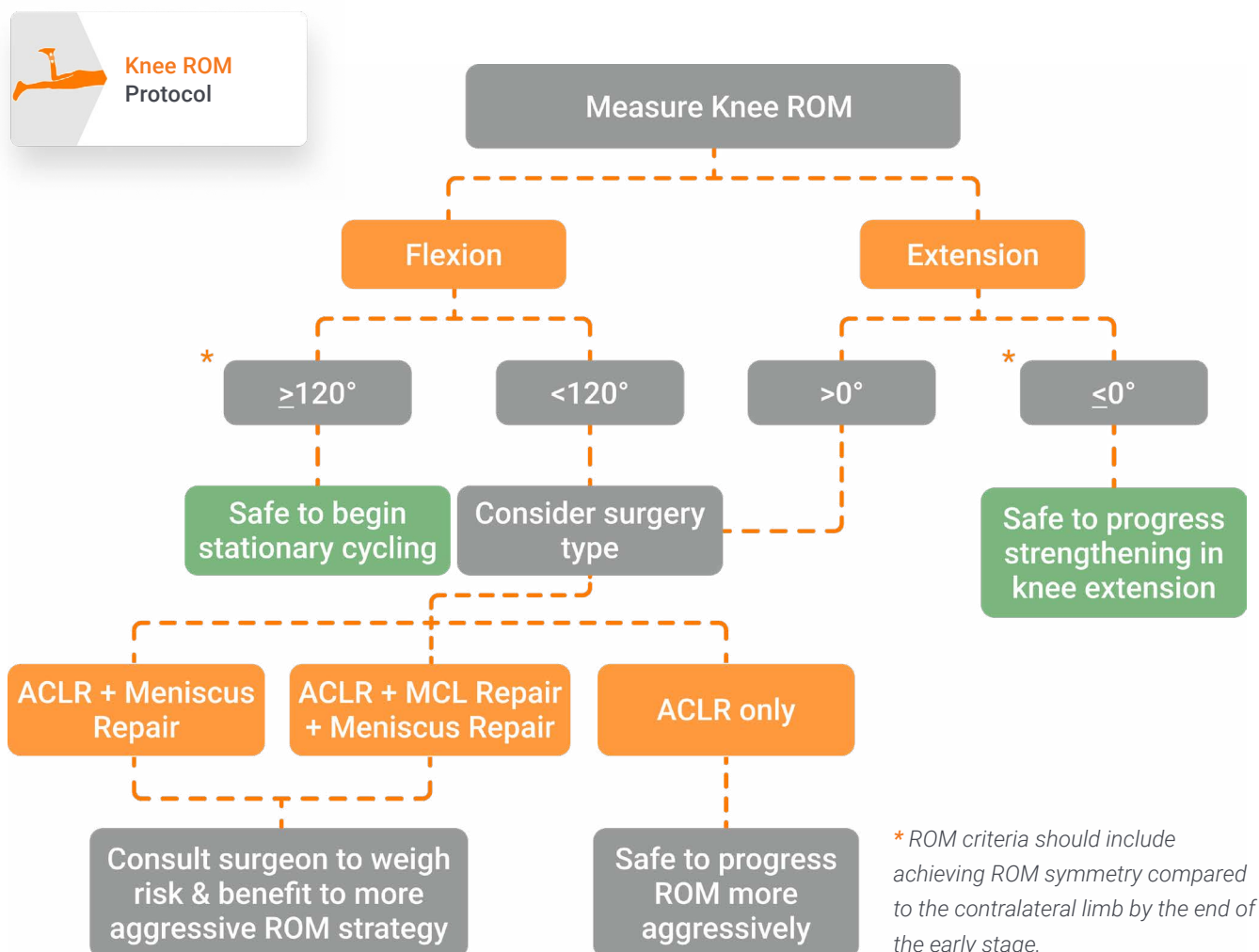
Knee flexion ROM is also important to monitor. Angles $>120^\circ$ should be an initial early-phase target to commence cycling, followed by achieving 95% of the uninvolved limb's ROM for return to running and full flexion ROM before returning to sport.

The type of surgical procedure should be considered and may require caution with early ROM (e.g., meniscal repair); therefore, flexion ROM trajectory recovery may be slower depending on the graft type and inclusion of other surgical procedures such as lateral extra-articular tenodesis (LET).



Early ROM Decision Tree

2-5 weeks post ACLR

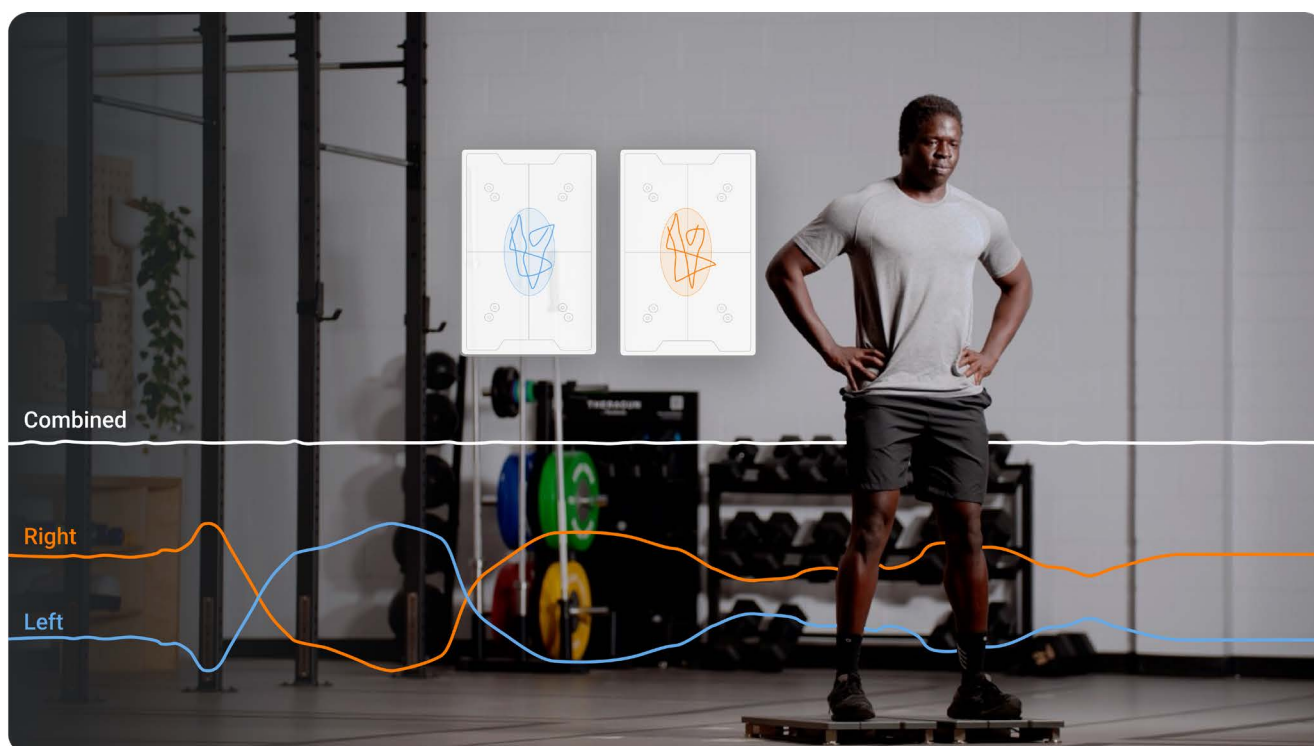


Movement Assessment in Early Rehabilitation

In addition to the established early-phase clinical goals of minimizing swelling, restoring ROM and regaining activation of key muscle groups, it is also important to re-establish movement quality and neuromuscular control during activities of daily living.

Standing **Weight Distribution**

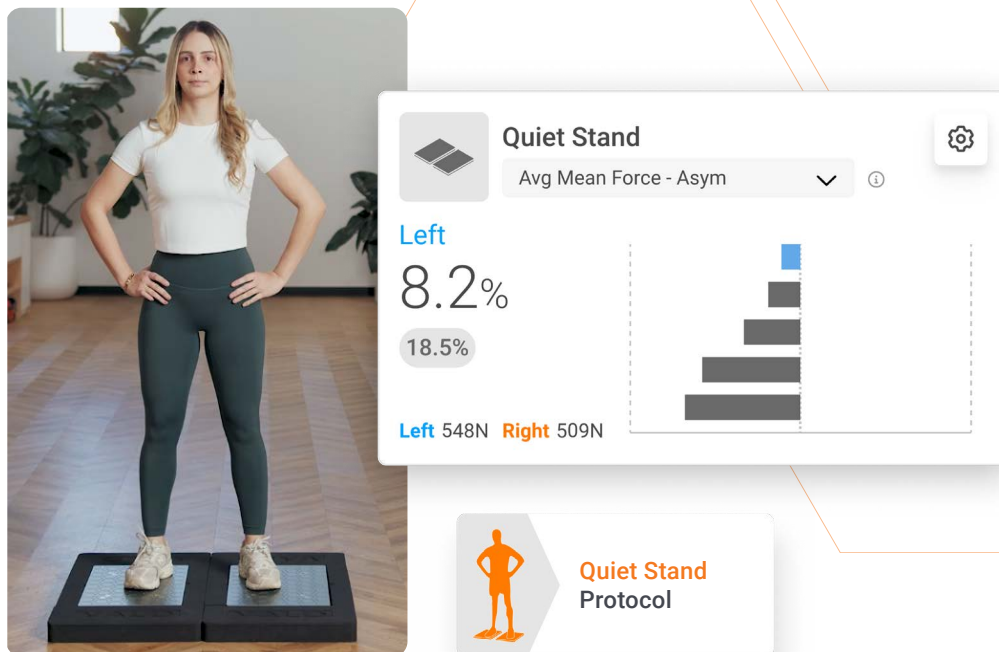
Early in the postoperative period, it is common for individuals to alter their weight distribution (50% on each leg representing perfect symmetry) to “offload” the reconstructed side.



Quiet stand assessment performed on ForceDecks, showing both center of pressure (CoP) traces as well as left and right force data for balance and weight shifting, respectively.

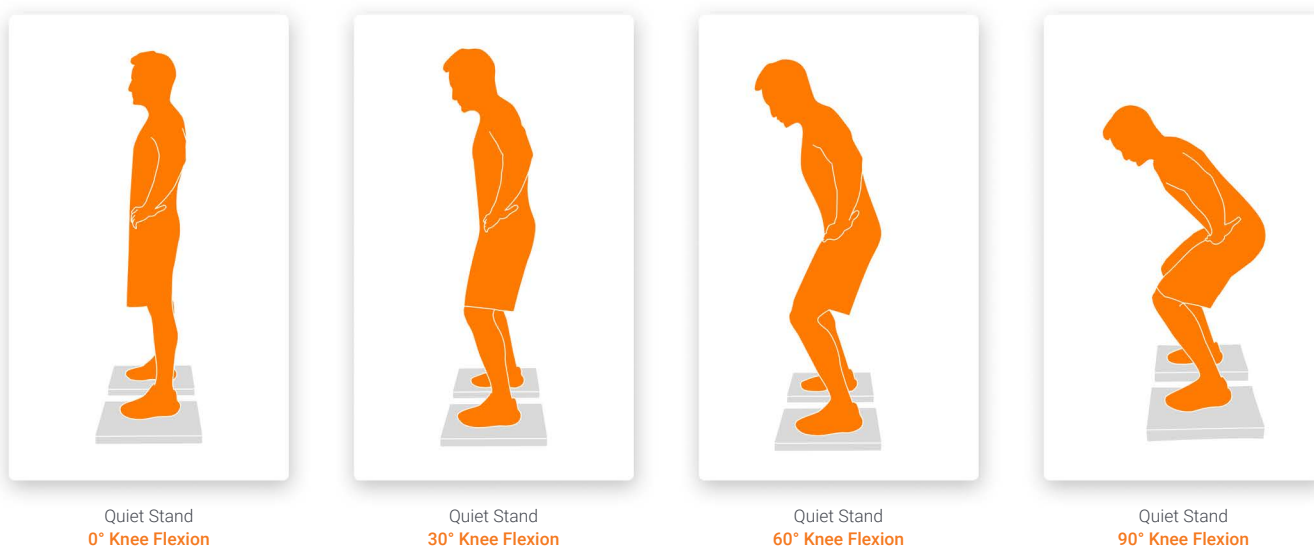
Significant differences compared to individuals without a history of ACLR can be expected during the initial weeks, with the percentage of weight accepted by the injured side gradually increasing and normalizing by approximately six weeks. Given this timeframe, monitoring this progress on a weekly basis may be beneficial.

...the percentage of weight accepted by the injured side [during the quiet stand will] **gradually increase and normalize by approximately six weeks.**



The assessment can be completed at a range of knee joint angles (0°, 30°, 60° and 90°).

Often, differences are magnified at 90° of knee flexion, showing strong relationships to quadriceps strength in the early phases of rehabilitation.



Single Leg Balance (Quiet Stand)

Postural instability and reduced balance are common following ACLR, with research indicating increased sway, higher velocity and changes in CoP.

These alterations in postural control are difficult to detect visually but can remain for up to one year post-surgery. Measurement in the anteroposterior and mediolateral planes is especially encouraged.

...alterations in postural control variables can **remain for up to one year post-surgery.**

Criteria for implementing unilateral balance assessment are not well-established in the literature; however, the following are important criteria to ensure safety when performing single leg balance tasks:

- ≤ 0 degrees of knee extension ROM
- No quadriceps lag (e.g., unintentional knee flexion during a straight leg raise)
- Discharged brace from gait training



Expert Tip:

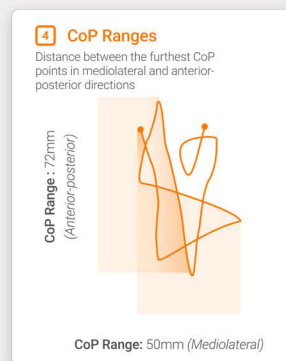
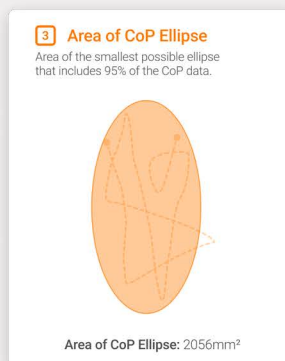
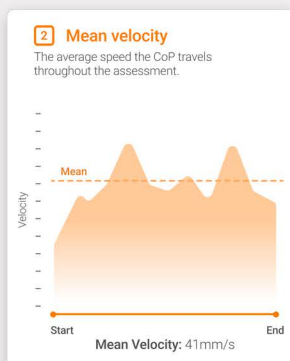
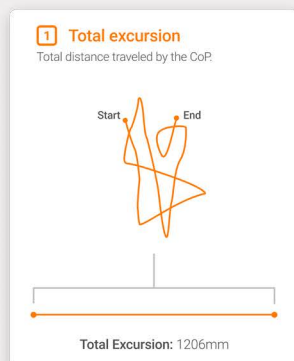
When using the single leg stance, it is important to instruct individuals to stand as still as possible and use an area where there are no external distractions or noises.

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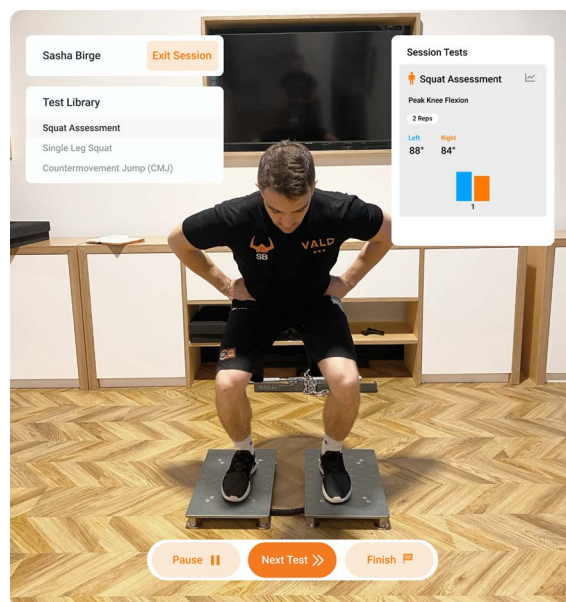
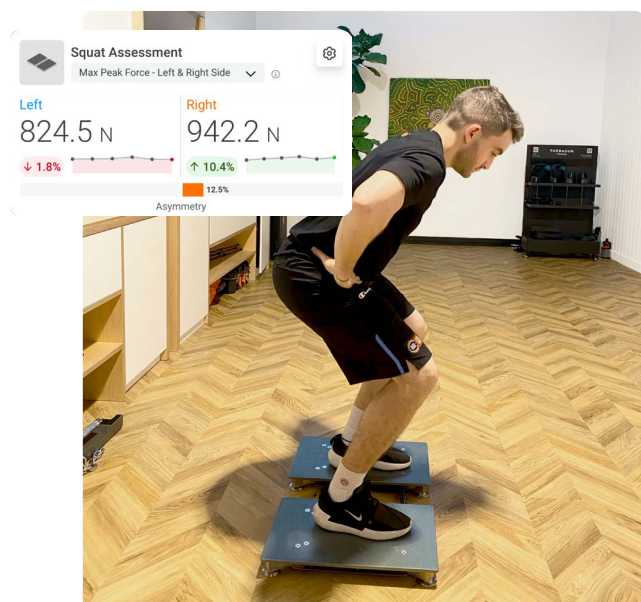


Single Leg Stand Protocol



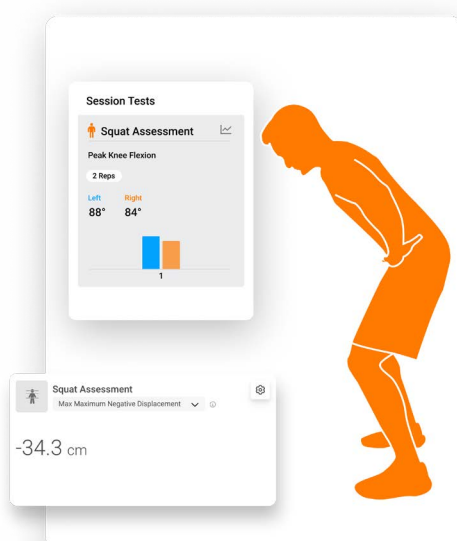
Squat Assessment

Once standing weight distribution has normalized close to symmetry (e.g., 5% asymmetry), the assessment can be progressed to evaluate the individual's lower limb dynamic force distribution both eccentrically and concentrically during a squat pattern.

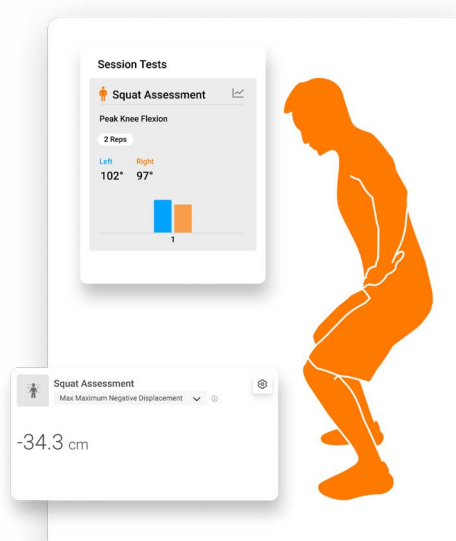


Following ACL injury or surgery, it is common to display inter-limb asymmetry in kinetics and kinematics during a bilateral squat.

Motion analysis (e.g., using [HumanTrak](#)) can identify movement strategy deficits, whereby athletes attempt to “offload” the force requirement of the involved knee.



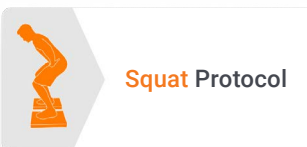
Squat Assessment
Greater Hip and Trunk Flexion
(Less Quadricep Control)



Squat Assessment
Less Hip and Trunk Flexion
(More Quadricep Control)

Asymmetry in ground reaction forces (GRFs) is common and can be readily assessed using force plates. These deficits affect landing mechanics and should be addressed before progressing to landing movements that involve higher GRFs.

Velocity of movement can also be assessed as a marker of confidence, as athletes will typically achieve a faster eccentric peak velocity as they progress in their rehabilitation.



...athletes will typically **achieve a faster eccentric peak velocity** as they progress in their rehabilitation.

When interpreting squat assessment data, note that asymmetries can stem from altered motor control, not solely limb capacity limitations. This can lead to chronic underloading of the injured leg during bilateral exercises.



However, asymmetries in lower extremity exercise can be acutely corrected via the use of real-time feedback. This approach can be effectively performed using motion analysis and force plates, offering biofeedback to enhance motor learning and correction of this learned pattern of non-use.

In addition, targeted strategies such as providing constraints (e.g., a one-sided heel elevation) and exercise modification (use of split-stance drills) can be effective.

It is important to reassess frequently to ensure that the chosen intervention is working to normalize between-limb symmetry.

Force plates, such as [ForceDecks](#), allow practitioners to place objects (e.g., a heel lift) on the plates before zeroing, allowing accurate data to be collected under different environments.

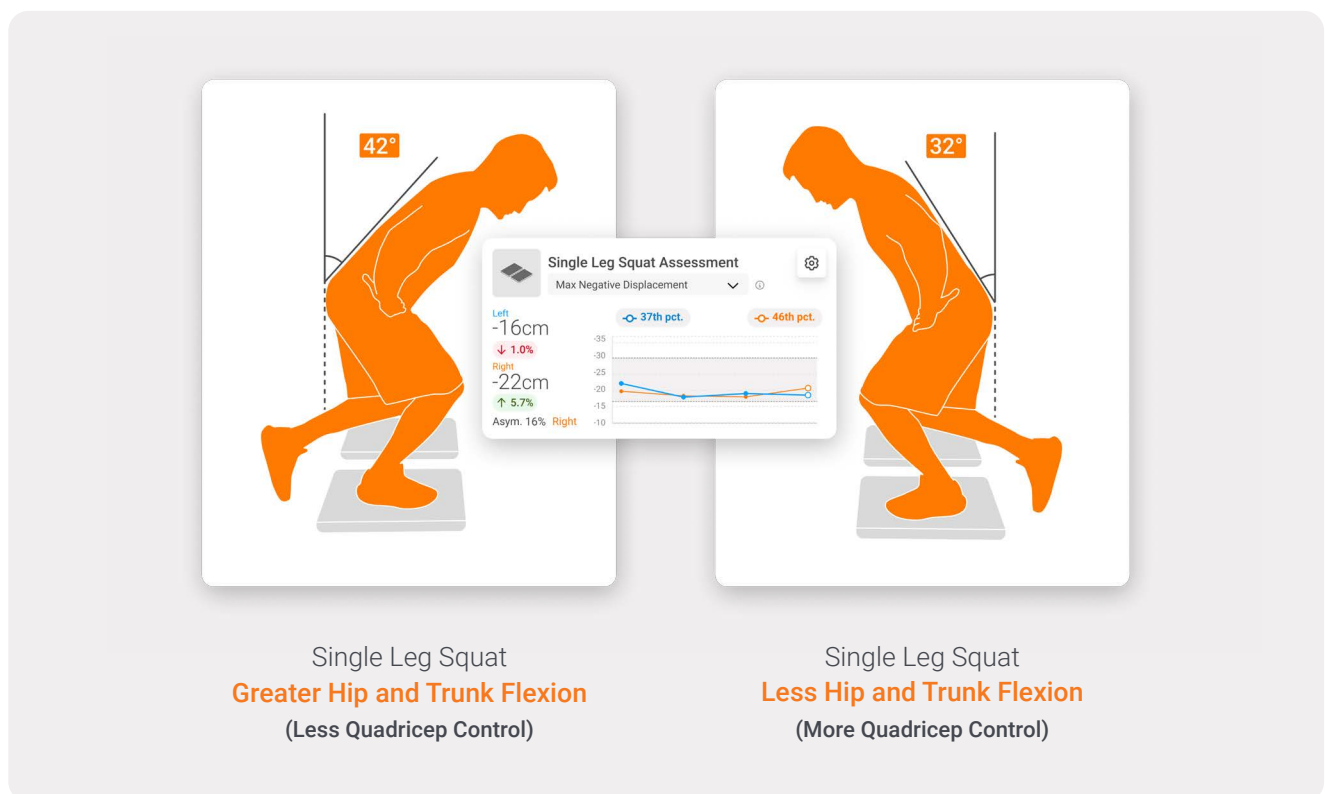
Single Leg Squat Assessment

The single leg squat is a more advanced early-phase assessment that isolates each leg to identify unilateral weaknesses. Compared to a bilateral squat, a single leg squat requires greater force production due to the need to support more of your body weight (BW) above your hips. The narrower base of support also challenges stability.

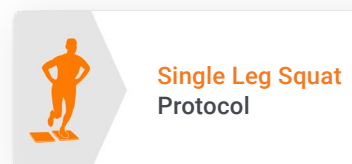
Following ACL injury, intralimb compensation would be expected in the early stages of rehabilitation. Such compensations often manifest in a similar depth of total movement (center of mass (CoM) displacement) with reductions in knee flexion, therefore decreasing the demand on the quadriceps.

In the presence of a quadriceps strength deficit, athletes will likely increase the amount of hip flexion and while decreasing ankle dorsiflexion to achieve the required squat depth.

In the presence of a quadriceps strength deficit, athletes will likely **increase the amount of hip flexion and decrease ankle dorsiflexion** to achieve the required squat depth.

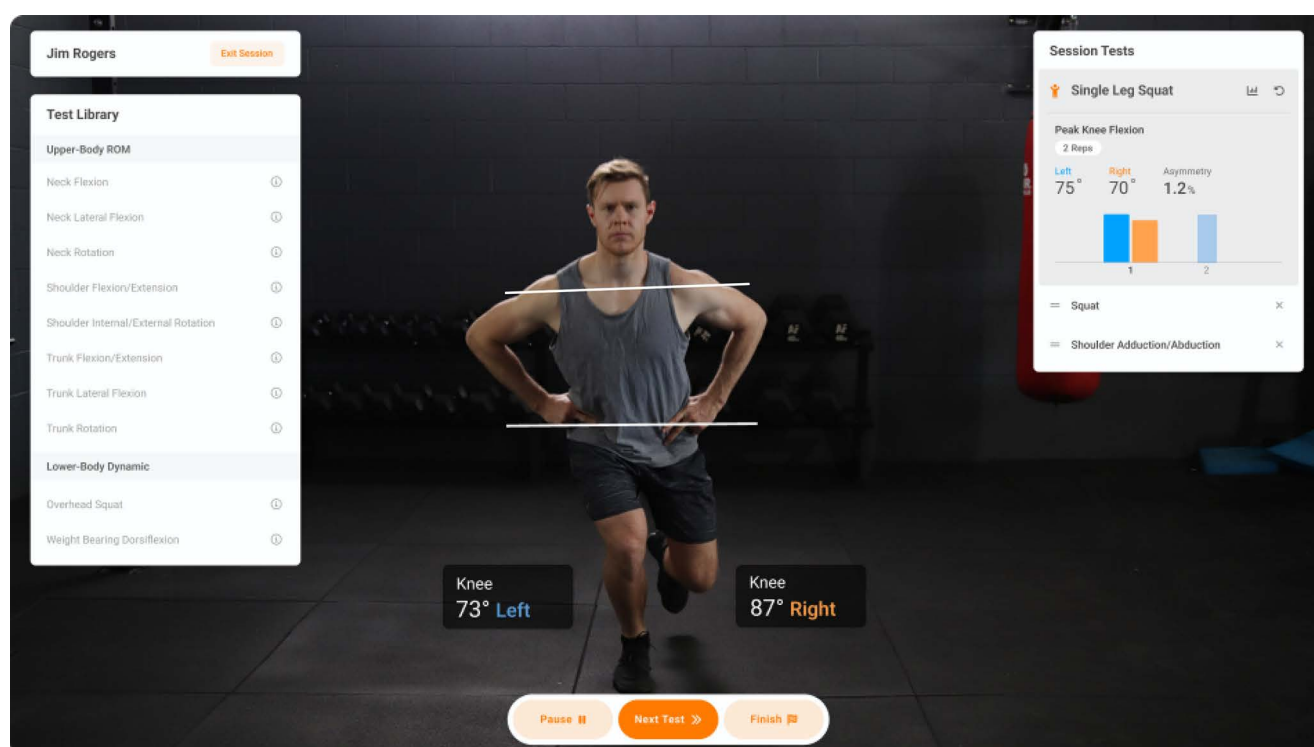


Movement strategy changes can be identified using motion analysis and are important to monitor longitudinally to ensure the functional capacity of the knee and active ROM is restored and reflected in movement strategy.



Specific kinematic deficits, such as knee flexion angle, appear to be particularly sensitive in the first few months post-surgery, with individuals displaying significantly less knee flexion compared to controls without a history of ACL injury.

Monitoring knee flexion angle during the 1–6-month period provides an effective metric for assessing an athlete's recovery progress. However, beyond this time point, force production markers may be more sensitive.



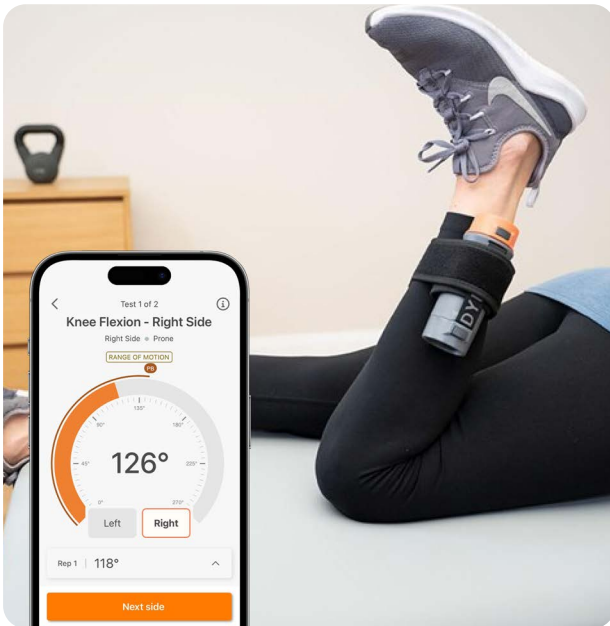
...knee flexion angle during the 1–6-month period provides an effective metric for assessing an athlete's recovery progress...**beyond this time point, force production markers may be more sensitive.**

The single leg squat can also act as an effective movement screen. If an athlete exhibits limited knee flexion ($<90^\circ$) at six months post-surgery, this may be deemed atypical. Using these insights, practitioners can delve deeper to understand why this deficit is present.

For example, the issue could be due to a loss of strength around the hip, knee or ankle, and this could be determined by employing joint-specific tests.

...if kinetic factors are ruled out, we can infer that the problem may be primarily kinematic (or, in chronic or complex cases, psychosocial) and subsequently focus on coaching strategies to enhance active knee flexion ROM.

Alternatively, if kinetic factors are ruled out, we can infer that the problem may be primarily kinematic (or, in chronic or complex cases, psychosocial) and subsequently focus on coaching strategies to enhance active knee flexion ROM. This approach ensures a more targeted and effective intervention using data-informed feedback.



It should be noted that a history of additional meniscus surgery is likely to result in **further reductions** in knee flexion ROM compared to those who have just undergone ACLR alone.

Furthermore, surgical procedures that have become more frequent, including LET, appear to also reduce the knee flexion ROM during a single leg squat.

Although this may not directly influence the treatment plan, it can offer valuable insights when profiling athletes, tracking their recovery progress and setting realistic rehabilitation goals.

Step Down (Land and Hold)

The step down is a functional assessment used to monitor movement control. Task completion time can be recorded and has been shown to be significantly slower during the **earlier stages** of rehabilitation; however, peak drop landing force is the main metric to measure.

Practitioners looking to increase the depth of their data can calculate the impact index (peak vertical impact force / BW * 100), which provides the peak impact force relative to BW.

Trials should be recorded where the involved (ACL) leg is responsible for the step up/down, and the uninvolved side contacts the force plate and vice versa.

When stepping down from the box using the involved limb, individuals typically display less control, resulting in faster and harder foot contacts.

Impact Index Calculator

Anterior Step Down Assessment

Peak Impact Force (N)

÷

Bodyweight (N)

×

Percent Normalization

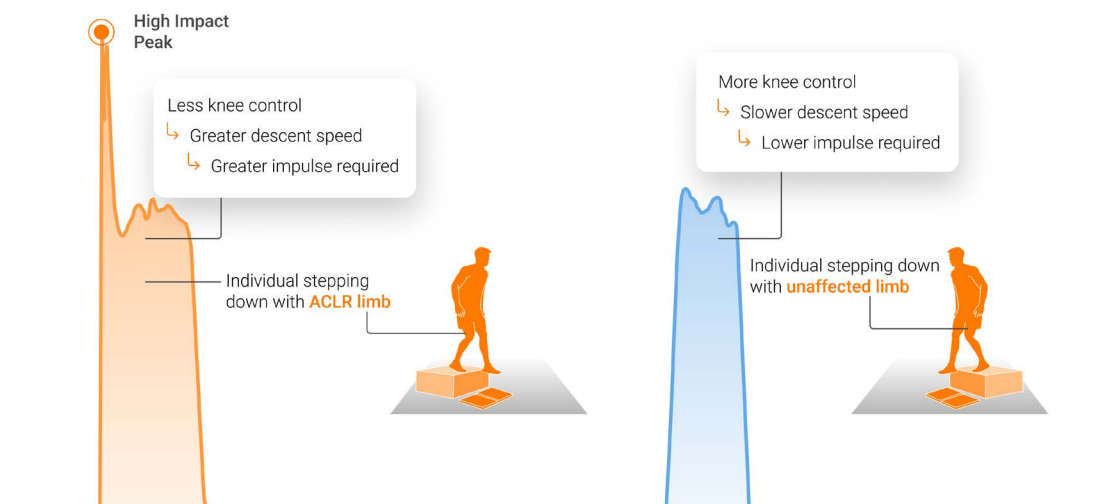
100

=

Impact Index

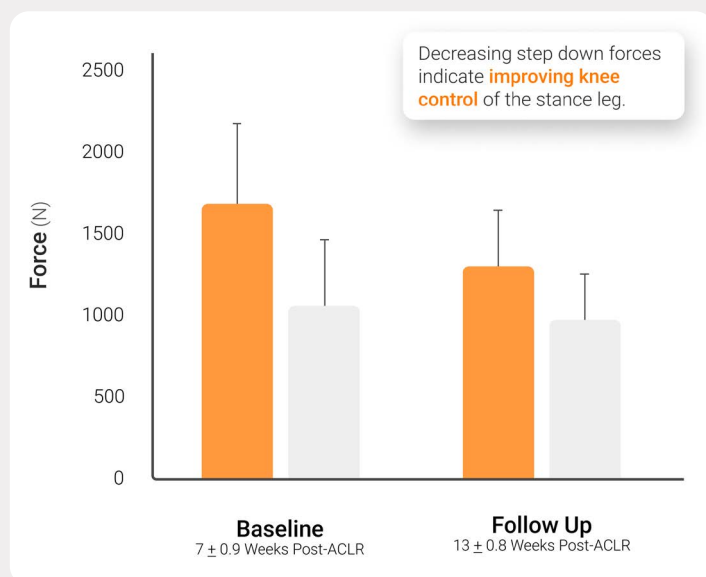
When stepping down from the box using the involved limb, **individuals typically display less control, resulting in faster and harder foot contacts.**

This can be seen in a force-time trace, where a pronounced higher peak vertical force is evident. This strategy appears to be more pronounced in the earlier phases of rehabilitation but improves over time.



Step down force example noting the impact peak and higher GRF when stepping down using the ACL reconstructed leg to control the movement.

A typical reduction of peak GRF of ~20% across a 6-week period can be expected, corresponding with increases in limb symmetry index (LSI) from 65-75%. This pattern is related to quadriceps strength, with nearly half of the variance in impact index explained.

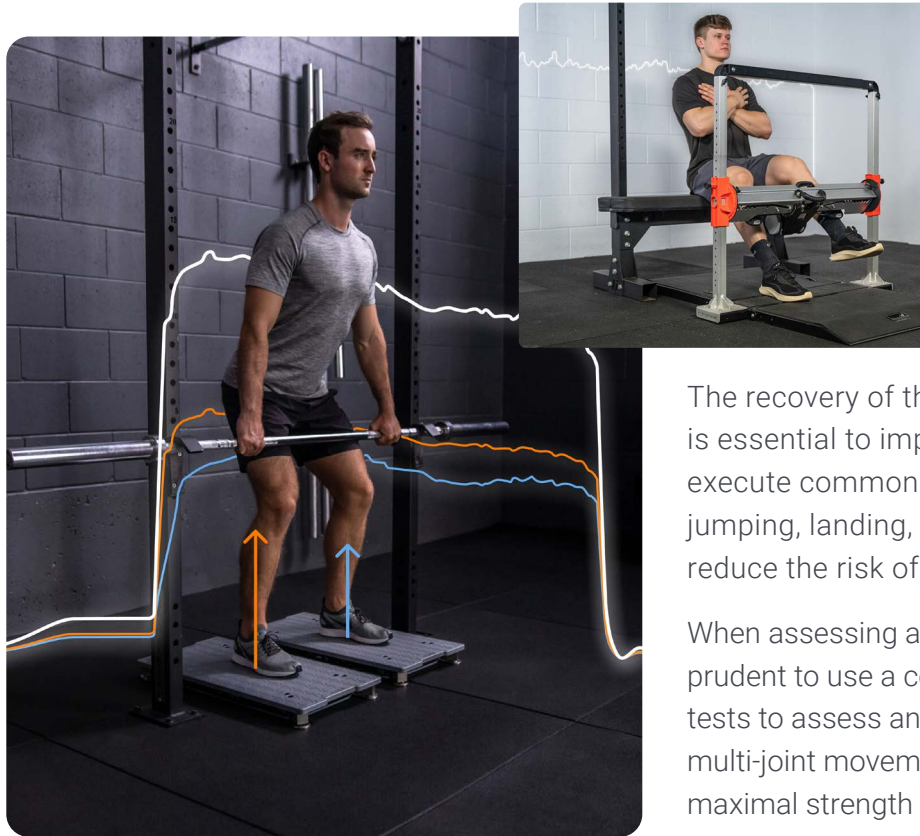


Observed changes in step down peak force during rehabilitation.
Note: a reduction in peak GRF indicates improved step-down control.

Want to learn more about movement analysis? Check out our [Practitioner's Guide to Force Plate Series](#).

Optimizing Strength Assessment

After ACLR, residual deficits are commonly reported in maximal and explosive strength (i.e., RFD).



The recovery of these key physical capacities is essential to improve long-term knee health, execute common athletic skills such as jumping, landing, sprinting or cutting, and reduce the risk of future injuries.

When assessing athletes following ACLR, it is prudent to use a combination of single-joint tests to assess and restore specific deficits and multi-joint movements to evaluate the athlete's maximal strength capacity.

Quadriceps Strength Deficits

Deficits in knee extension strength are commonly observed in the ACL reconstructed limb throughout rehabilitation and at the time of return to sport.

...greater symmetry in quadriceps strength is associated with a **reduced risk of injury prior to return to sport.**

Although isokinetic dynamometry is indicated as the gold standard, inline dynamometry (for example, using **DynaMo** with a strap attachment) can provide an alternative, **reliable and valid** measurement during ACL rehabilitation.

When inline dynamometry is used, testing at different knee joint angles is warranted to better understand the athlete's strength profile through range.

Research has indicated that greater symmetry in quadriceps strength is associated with a **reduced risk** of injury prior to return to sport, and preoperative quadriceps strength is a significant predictor of knee function **two years after ACLR**.

Atypical landing mechanics are also often shown in those with **reduced quadriceps strength** following ACLR.

Therefore, precise monitoring of quadriceps strength is a key component of ACL rehabilitation.

Responsible for Restoring

- Bilateral and unilateral squat mechanics
- Typical landing kinetics
- Isometric squat performance

Predicts

- Functional outcomes at 6 months
- Risk for second injury

Helps Improve

- Vertical jump performance
- Braking durations
- Eccentric peak velocity
- RSI-Mod

Lack of Strength Leads To

- Atypical landing mechanics
- Greater hip flexion during squatting
- Greater GRFs during step downs

The frequency of assessing strength may vary during different phases of recovery. In the presence of pain, consider testing on a weekly basis as progress occurs more rapidly due to changes in pain and reduced neuromuscular inhibition. As pain subsides and progress is driven more by physiological changes, less frequent testing may be suitable.



Knee Extension Protocols

In the presence of pain, consider testing on a weekly basis as **progress occurs more rapidly due to changes in pain and reduced neuromuscular inhibition.**

Be wary of testing too infrequently and not identifying when an inappropriate stimulus has been provided in a rehabilitation program. This has a downstream effect of delaying the time to reach benchmarks to satisfy criteria-based progression.

Expert Tip:

Measuring torque is beneficial because dividing by the moment arm normalizes values, enabling comparisons across different limb lengths and prompting clinicians to convert peak forces into peak torque for accurate performance comparisons to research.



Paul Read

PhD, ASCC, CSCS*D, PGCE

moment
arm (m)

force (N)

$$\text{force (N)} \times \text{moment arm (m)} = \text{torque (Nm)}$$

Inline dynamometry, such as with the DynaMo, is a valid and reliable alternative to isokinetic dynamometry.

Hamstring Muscles

Strength Deficits

Lower peak [knee flexion strength](#) has been displayed at the time of return to sport in comparison to healthy controls.

These deficits appear to be more evident when a semitendinosus-gracilis (hamstring) graft is used and is more noticeable at deeper knee flexion angles (i.e., >75° knee flexion).



...deficits appear to be **more evident [with] a semitendinosus-gracilis graft...**



The hamstrings are important during forceful dynamic movements, as hamstring coactivation provides dynamic knee joint stabilization and [prevents excessive ACL shear forces](#), reducing the load on the ACL.

Furthermore, a history of severe knee injury (including ACL injury) has been shown to increase the risk of a [future hamstring strain](#) injury.

Therefore, assessment of hamstring strength should be included as part of testing throughout rehabilitation.

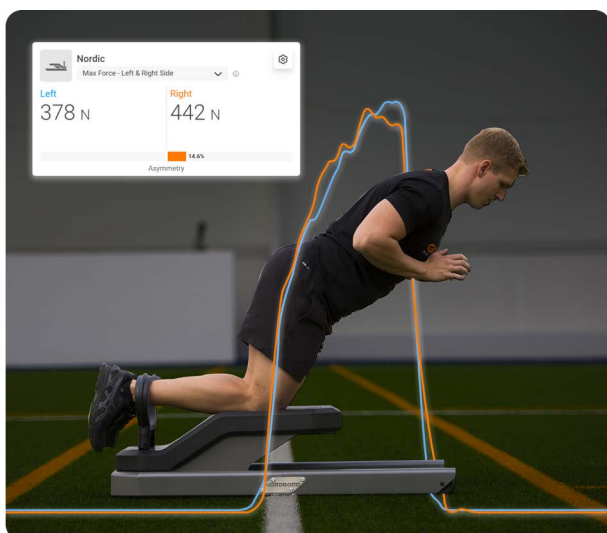
A variety of contraction modes should be included as part of a comprehensive testing approach. Architectural changes, including [shorter hamstring muscle fascicles](#), have been shown following ACLR with a [hamstring](#) graft.

Eccentric muscle actions are a key component of sprinting during the swing phase, and graft failure may also occur when the hamstring muscle actively lengthens to [resist anterior tibial translation](#).

Therefore, in addition to isometric tests performed using the DynaMo, [ForceFrame](#) or [NordBord](#), eccentric actions should also be examined.

Team sport athletes with a history of ACL injury (range 1-9 years) display lower peak eccentric knee flexor strength measured using the NordBord compared to the contralateral limb, and deficits appear to be larger in [females](#) (mean difference -53.7N) [than males](#) (-43.9N).

Due to consistent observations of lower eccentric strength...**ongoing assessment of eccentric hamstring strength is recommended.**



More recent research has also shown female team sport athletes produced [less eccentric knee flexor torque](#) on their involved side (5.86 vs. 6.59N.m.kg⁻¹) with a mean asymmetry of 6.3%.

Due to consistent observations of lower eccentric strength in athletes with a history of ACLR, even up to 10 years following surgery, ongoing regular assessment of eccentric hamstring strength is recommended.

Optimizing Knee Extension/Flexion Strength Assessment

Testing knee extension and flexion strength is a cornerstone of any quality rehabilitation plan. Below are six primary considerations for optimizing knee strength assessment.

It may also be prudent to delay testing until 6-12 weeks (depending on contraction mode and athlete presenting symptoms) to allow adequate healing of the harvested graft site.

Strength deficits post-ACLR are often **between 50-80° knee flexion.**



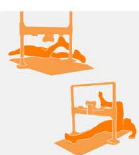
Measurement

Asymmetry alone may mask deficits in capacity. Aim for knee extension torque $>3.0 \text{ N}\cdot\text{m}\cdot\text{kg}^{-1}$ as a return-to-sport baseline. Calculate peak torque (Nm) by multiplying force (N) by the moment arm, then normalize by body mass (BM) (Nm/kg).



Screening

Pre-injury benchmarks are more relevant than symmetry alone. Athletes often meet pass criteria ($>90\%$ LSI) post-rehabilitation but remain below pre-injury performance.



Position

Angle-specific testing reveals deficits not captured by peak torque alone. Deficits post-ACLR are often between 50-80° knee flexion. Early rehabilitation may limit testing to 90° and 60°. Hamstring deficits are more pronounced at greater knee flexion angles, requiring inner-range strength testing.



Data Management

A 60% ratio is typical but may be inflated on the injured side due to reduced quadriceps torque. Test multiple angles to provide accurate dynamic knee function data, as hamstring:quadriceps ratios vary widely across ROM.



Surgical Considerations

Strength deficits depend on graft type (e.g., hamstring grafts impair flexors, bone-patellar tendon-bone (BPTB) affects extensors). Delay testing 6-12 weeks post-op to allow graft site healing.



Expert Tip:

Set straps and plinth $\sim 10^\circ$ below the intended angle to account for knee angle changes during maximal effort. This protects the graft during early rehabilitation.

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Minimizing **Atrophy** and **Strength Loss**

Strength deficits, especially in the hamstrings and quadriceps, are expected but trainable. However, starting rehabilitation early, ideally preoperatively, minimizes strength loss.

Early focus on quadriceps activation and torque production is critical, as knee extension torque at 6 weeks post-surgery predicts functional outcomes at 6 months.

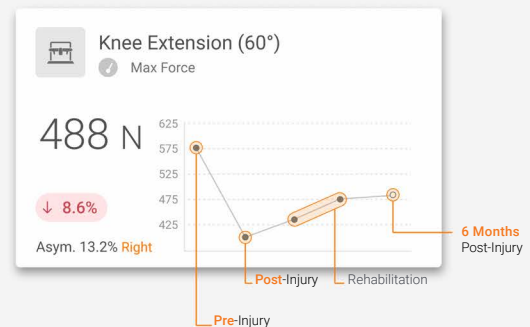
Early focus on quadriceps activation and torque production is critical, as knee extension torque at 6 weeks post-surgery predicts functional outcomes at 6 months.

Effects of **Early Rehabilitation**



- ✓ Greater restoration of quadriceps strength
- ✓ Faster recovery to pre-injury strength levels
- ✓ Earlier progression to advanced training
- ✓ Maintained strength levels during return to play

Effects of **Delayed Rehabilitation**



- ✗ Lower restoration of quadriceps strength
- ✗ Slower recovery to pre-injury strength levels
- ✗ Later progression into advanced training
- ✗ Strength levels decline during return to play

This often presents a challenge as heavy load resistance training is contraindicated in the early phases of ACL rehabilitation.

More recent approaches have included strategies to provide a potent stimulus to the muscle while keeping mechanical stress low. An example of this is via blood flow restriction (BFR) training.

Blood Flow Restriction (BFR) Training



Stephen Patterson

St Mary's University Twickenham London

Applied Exercise Physiology & Performance

BFR training, popularized by pneumatic tourniquet systems, stimulates muscle growth and strength at low loads, making it a valuable rehabilitation tool. It allows early hypertrophy and strength gains, offering a safe alternative to heavy lifting – especially for ACL patients unable to tolerate high loads.

BFR is a training method that has become more popular in recent years, particularly with the introduction of automatic pneumatic tourniquet systems (such as [AirBands](#)).



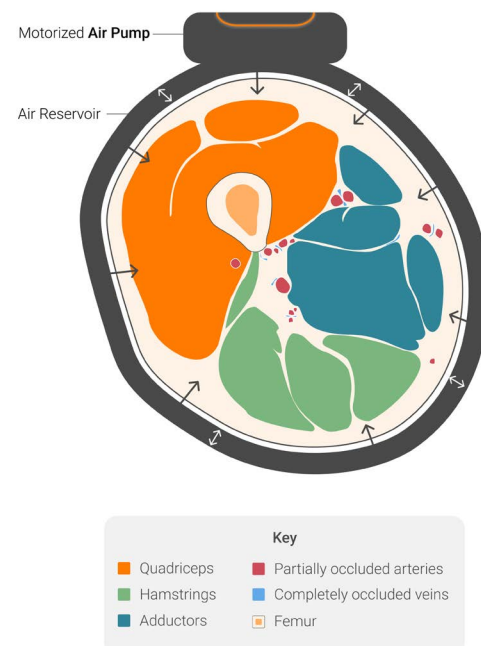
BFR partially restricts arterial inflow and fully restricts venous outflow in active muscles during exercise, resulting in hypoxia within the muscle.

In the context of ACL rehabilitation, the cuff is applied to the most proximal aspect of the lower limbs.

The low-load nature of BFR training and its ability to stimulate muscle hypertrophy and subsequent strength gains make it a powerful clinical rehabilitation tool.

BFR can, therefore, be used early in rehabilitation, offering an alternative to heavy-load resistance training.

This is particularly important for ACL-reconstructed patients who need to restore muscle hypertrophy and strength but cannot yet safely tolerate heavy loading of the musculoskeletal system.



[BFR] is particularly important for ACL-reconstructed patients who need to **restore muscle hypertrophy and strength but cannot yet safely tolerate heavy loading** of the musculoskeletal system.

Augmenting Rehabilitation with BFR

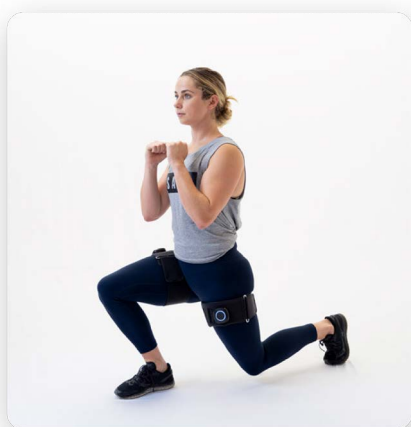
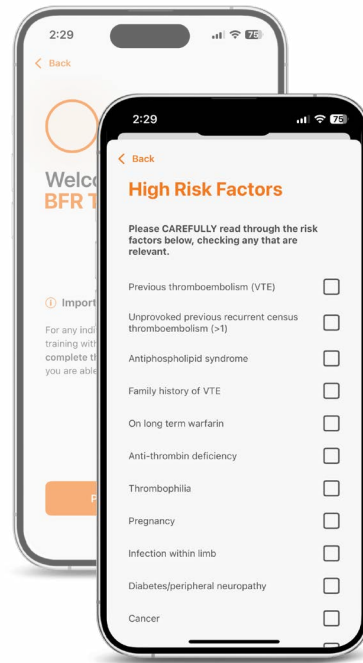
Passive BFR and low-load BFR exercise can be used during postoperative rehabilitation to combat muscle atrophy and strength loss.

Passive BFR (the application of BFR without exercise) can be used early and creates cellular swelling, which increases muscle protein synthesis and suppresses protein breakdown.

This phase can begin as little as a few days post-surgery with controlled inflammation, pain and swelling as long as patients have passed a risk assessment questionnaire.

In addition, voluntary isometric contractions during BFR may increase metabolic stress and cell swelling levels above passive BFR and contribute to muscle hypertrophy, acting as preparation for subsequent low-load BFR exercise.

As ROM is restored and gait is normalized, low-load resistance with BFR can be introduced to **accelerate muscle hypertrophy and improve strength.**



As ROM is restored and gait is normalized, low-load resistance with BFR can be introduced to accelerate muscle hypertrophy and improve strength.

During BFR resistance exercise, early preferential recruitment of type II fast-twitch fibers at low loads is also thought to be an important mechanism behind strength adaptations, potentially helping to combat arthrogenic inhibition.

The use of neuromuscular electrical stimulation, both during and separate from BFR training, can also augment this process.

Implementing BFR Training

Research supports individualization, where BFR is prescribed as a percentage of “arterial limb occlusion pressure (LOP),” which represents the minimum pressure required for total [arterial occlusion](#).

A brief overview of the current consensus on BFR application during rest and exercise is provided in Table 1 based on the work of [Patterson et al. \(2019\)](#).

Criteria	Phase 1 (Passive BFR)	Phase 2 (Active BFR)
Frequency	1-2 times per day	2-3 times per week
Type	Small and large muscle groups	Small and large muscle groups
Load	N/A	20-40% of 1 repetition maximum
Time	5-minute intervals	5-10 minutes per exercise
Volume	3-5 sets	Option 1: 30, 15, 15, 15 (75 reps) Option 2: 2-4 sets to failure
Rest Between Sets	3-5 minutes	30-60 seconds
Pressure	Uncertain – higher pressures (70-100% LOP) may be needed	40-80% LOP
Restriction Form	Continuous	Continuous
Execution	N/A	Until concentric failure or when planned rep scheme is completed
Execution Speed	N/A	1-2 seconds per phase (e.g., concentric)

BFR has consistently been shown to improve strength and hypertrophy in the early stages of ACL rehabilitation.

However, practitioners are cautioned not to over-rely on BFR as a strengthening modality, as middle and late stages of rehabilitation may be best suited for heavier and faster loads to more adequately prepare for return to activity.

Rate of Force Development

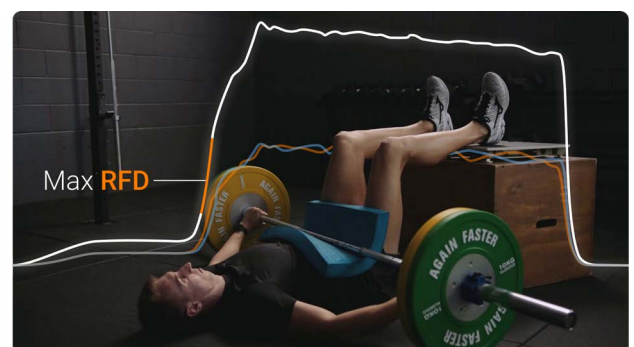
Impaired knee extension RFD has been shown following ACLR. Given ACL injuries occur shortly after ground contact (40-80ms), the ability of the quadriceps to generate force rapidly is important for optimizing lower extremity loading characteristics in hopping and jumping.



Knee flexor deficits after ACLR are also **typically more pronounced for RFD than maximal strength.**

Knee flexor deficits after ACLR are also typically more pronounced for RFD than maximal strength. Therefore, including knee extensor or flexor RFD assessments may be useful to comprehensively profile an athlete's explosive force production characteristics.

Research has also shown that this physical quality may take longer to restore post-surgery than maximal strength, further supporting the need for precise monitoring of specific adaptations.



Measuring RFD Reliably

To measure RFD, some important considerations are required. Although these are not specific to the knee (similar formats are used for upper extremity testing as well), they are especially pertinent for maintaining data accuracy.



Sampling Rate

Use devices with appropriate sampling frequency. A 1,000Hz (1,000 data points per second) sampling rate is generally recommended to quantify RFD. Technologies such as ForceDecks (1,000Hz) or DynaMo Max (1,200Hz) are therefore often preferred.



Test Selection

Use the appropriate test mode. Isometric assessments are recommended for the measurement of RFD, with substantial variability expected when RFD is measured during jump testing.



Cueing

Ensure representative cues are used. During an isometric assessment measuring RFD, athletes should be cued to push or pull "as fast as possible," as this physical quality is sensitive to the instructions given.



Test Separation

Use separate tests for maximal strength and RFD. Attempting to achieve maximal force and RFD within the same contraction may result in suboptimal measures of both parameters given.



Familiarization

Familiarize athletes prior to testing. This is important for all tests, but the specific cueing instructions highlighted above should be practiced.



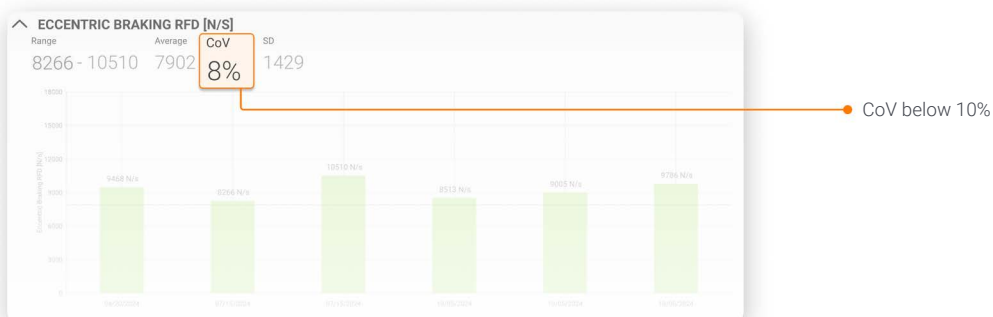
Degrees of Freedom

Single-joint tests with appropriate constraints may be more accurate than multi-joint tests with more degrees of freedom. Ensure athletes are fixed into position with straps where possible and prioritize single joint tests to minimize variability and maximize the sensitivity of between-limb differences.



Reliability

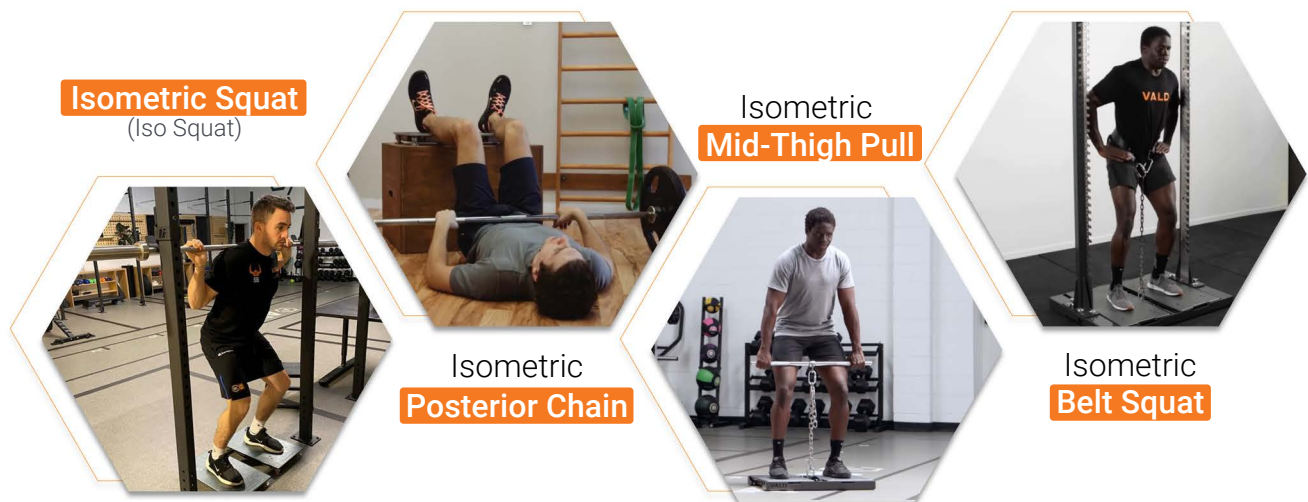
Check the reliability of your data. Examine the coefficient of variation (CoV) to determine acceptable within-trial variability. While no evidence-based thresholds exist, pragmatically aim to keep values <10%.



Measuring Strength in Compound Movements

Multi-joint strength characteristics (e.g., squat) display a heightened transfer to athletic performance, including jumping, sprinting and change of direction (COD).

It is not currently known if multi-joint strength or isometric strength characteristics measured during an Iso Squat or isometric mid-thigh pull (IMTP) differ in athletes following ACL surgery in comparison to healthy controls.



Isometric multi-joint tests measure general physical characteristics that underpin various aspects of sports performance; thus, it may be prudent to include them as part of a comprehensive strength profiling assessment.

They can also be used to identify trainable deficits, which may enhance readiness to re-perform following long-term injury.

The test selected will depend on what capacity you wish to measure optimally. In an Iso Squat,

higher force production is shown, whereas the IMTP displays a higher RFD.

An Iso Squat also increases the recruitment of the knee extensors, so it may be advised when testing athletes following ACLR due to the likelihood of residual quadriceps weakness.

Learn more about isometric testing in the [Practitioner's Guide to Isometrics](#).

An Iso Squat also increases the recruitment of the knee extensors, so it may be advised...due to the likelihood of residual quadriceps weakness...

Can Multi-Joint Tests Be Used to Enhance Specificity?

ACL injuries often occur with the knee near full extension. Therefore, measuring strength characteristics in positions that resemble sporting actions and positional injury mechanisms could be considered. An example of this is a Knee Iso-Push – part of the [Run-Specific Isometric Strength Test battery](#) developed by Alex Natera.



Alex Natera, PhD, MSc
Performance Science Manager at NSWIS

Creator of Run-Specific Isometric Strength Testing



During the Knee Iso-Push assessment, evaluating metrics like force production relative to BM and force at specific time points (e.g., 100ms and 200ms) can reveal limitations or qualities that might not be identified through other assessments.

...evaluating metrics like force production relative to BM and force at specific time points (e.g., 100ms and 200ms) **can reveal limitations or qualities that might not be identified through other assessments.**

It is recommended to pair this assessment with the supine Hip Iso-Push so that the force production characteristics of the posterior chain can also be measured.

Beyond the Knee

When considering the function of the knee joint, it is important to consider muscles that are proximal and distal as they play a role in dynamic stabilization.

This section details some key muscles to include as part of a holistic strength assessment.



Hip Abductors

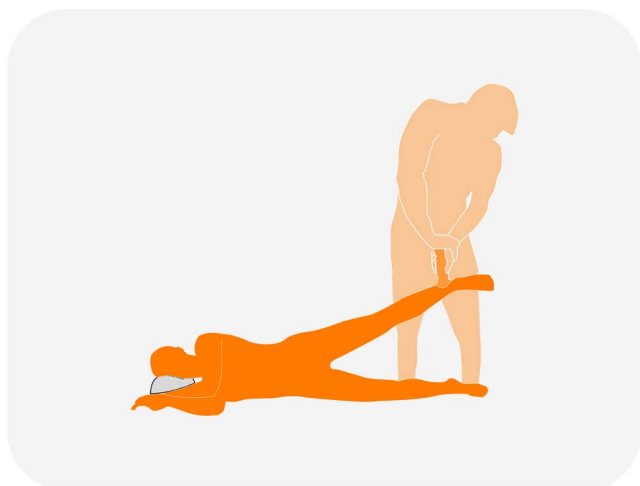
Hip abduction strength can be considered relevant when examining frontal plane movement activities, with increases in torque associated with [reduced knee valgus motion](#).

Furthermore, female team sport athletes with a history of ACLR have displayed [lower isometric hip abductor torque](#) on their involved side.

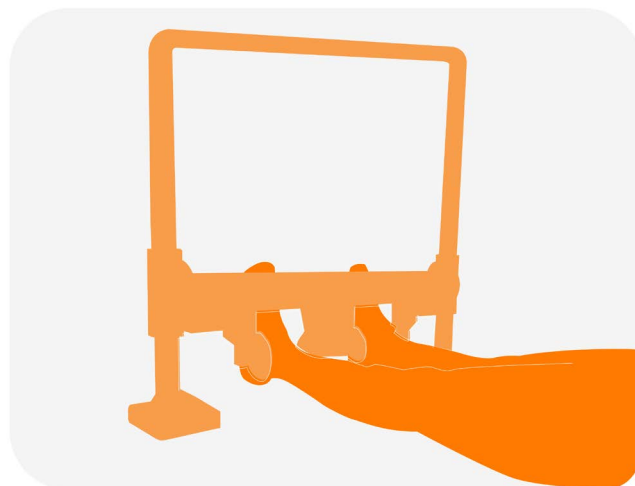
...female team sport athletes with a history of ACLR have displayed **lower isometric hip abductor torque on their involved side.**

Hip abduction eccentric strength can be assessed unilaterally using handheld dynamometry. Normative values between [2.2 and 3.0Nm/kg](#) have been reported in healthy male soccer players.

If using the supine long lever isometric abduction strength test, values of approximately [2.3Nm/kg](#) should be expected.



Eccentric **Hip Abduction**
(Handheld Dynamometry)

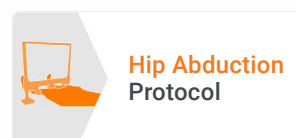


Isometric **Hip Abduction**
(Fixed-Frame Dynamometry)

Bilateral assessment of hip abduction in supine with straight legs has also been used in female team sport players with a [history of ACLR](#).

The involved side displayed lower isometric hip torque than the **contralateral side (1.9 vs. 2.04N.m.kg⁻¹).**

The involved side displayed lower isometric hip torque than the contralateral side (1.9 vs. 2.04N.m.kg⁻¹). Hip torque asymmetry was also [included in a predictive model](#) that best discriminated between players with and without a history of ACL injury.



Ankle Plantar Flexors

While limited data are available to indicate weakness in these muscles post-ACL injury or reconstruction, assessment of the ankle plantar flexors is encouraged as the soleus has an important antagonist role in modulating mechanical load around the knee.

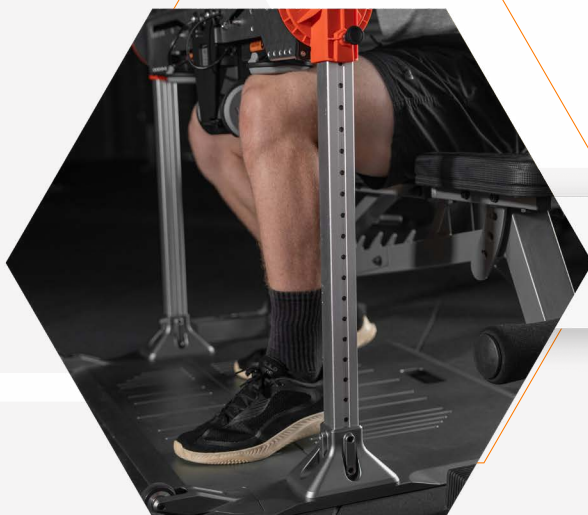
This may be especially important for those with a hamstring graft due to the increased likelihood of knee flexor deficits.

The soleus can produce posterior shear force to protect the ACL during cutting and landing activities. It has been observed that these forces are up to **32% of those produced by the hamstrings**. Deficits in ankle plantar flexion could also impact propulsive impulse during running and jumping.

The **soleus can produce posterior shear force to protect the ACL** during cutting and landing activities.

Seated ankle plantar flexion can be measured using the ForceFrame or with an isometric testing rig and force plates, such as ForceDecks, which are capable of a range of specific isometric calf tests.

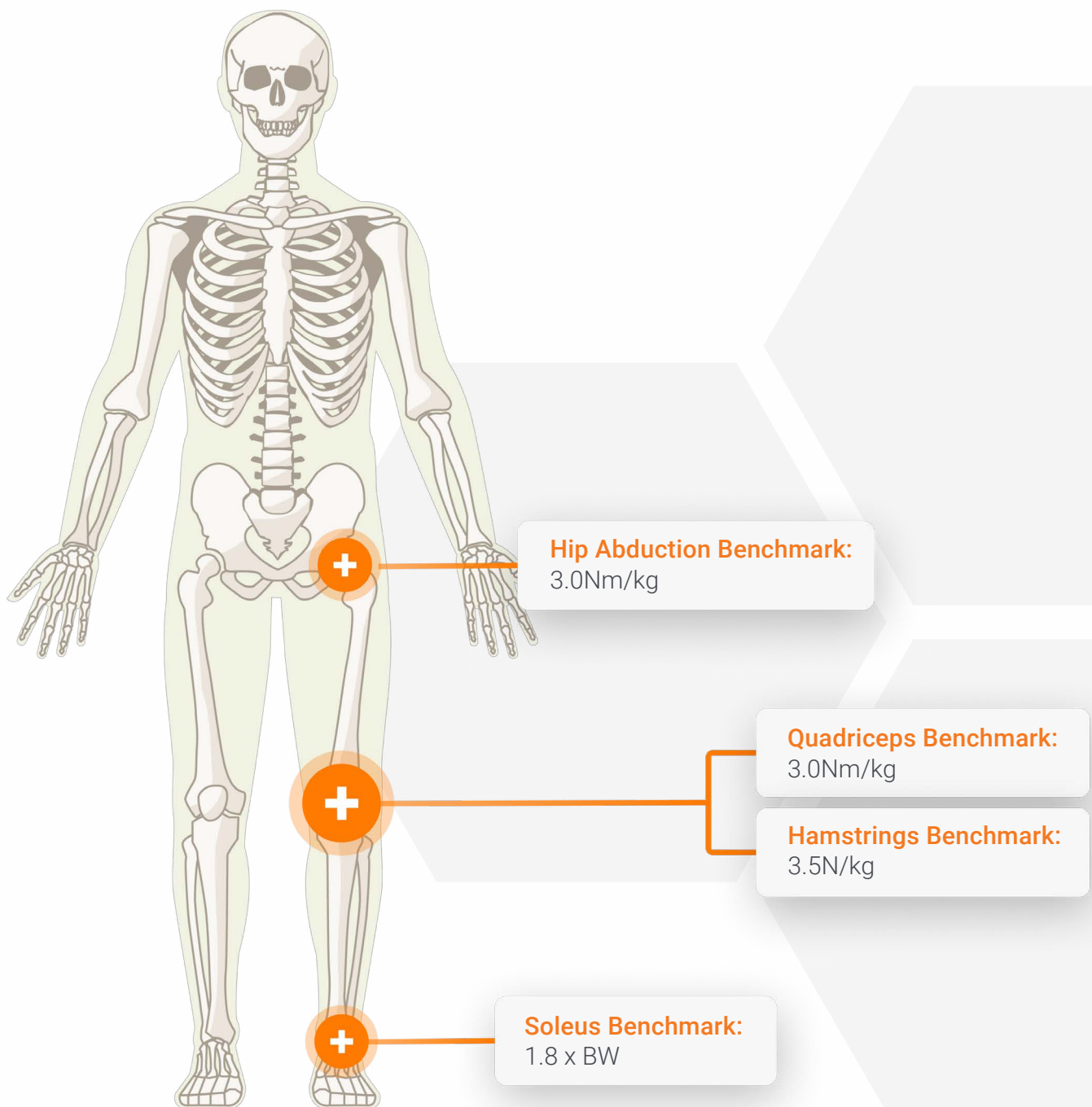
Isometric net force **>1.8 x BW** in a neutral position is a recommended target to meet the values of healthy athletes. Due to the high force demand for the plantar flexors during running, it is also recommended to include the **Ankle Iso-Push** assessment.



Soleus Benchmark:
1.8 x BW

Strength **Summary**

These are the following benchmarks that athletes and practitioners can aim for throughout the course of ACL rehabilitation.



Regional assessment benchmarks for isometric testing in three key locations for ACL rehabilitation.

Jump Assessments

Why We Use Vertical Jumping in ACL Rehabilitation

Jumping and landing tests such as the countermovement jump (CMJ) and single leg jump (SLJ) require rapid eccentric and concentric actions inherent to many athletic movements across multiple sports. The CMJ is a bilateral support task, allowing athletes to adopt between-limb compensation strategies in the presence of residual deficits after injury.

Greater asymmetry has also been shown in jumping compared with COD tasks [nine months after ACLR](#), supporting the use of jump tests to monitor progress during rehabilitation.

Greater asymmetry has also been shown in **jumping compared with COD tasks nine months after ACLR...**

Importantly, jumping [vertically](#) involves greater work to be performed at the knee during the propulsive phase compared to a horizontal jump. As a result, vertical jumps may be able to better identify deficits in knee function.

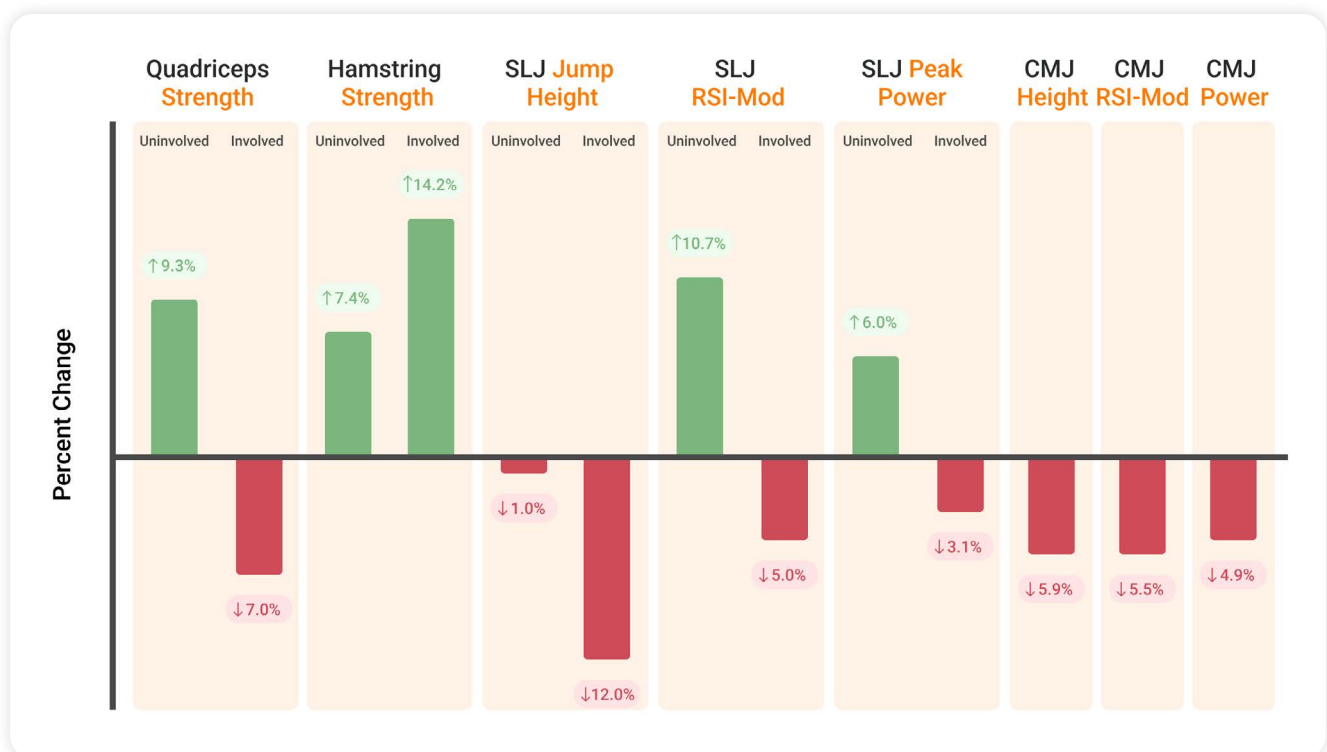


Vertical GRF measured via force plate technology is associated with [knee joint moments](#), indicating their viability as a surrogate for assessing compensation strategies in knee kinetics during different phases of the CMJ.



Common Residual Deficits

Significant improvements in jump height and power between six and nine months post-ACLR are expected; however, it is not uncommon to observe no further improvements after this point.



Adapted from [Maestroni et al. \(2022\)](#) showing CMJ, isokinetic and SLJ percentage change values comparing pre-injury to post-rehabilitation.

Similarly, comparisons to pre-injury benchmarks have shown that soccer players display residual deficits in jump height, power and reactive strength index-modified (RSI-Mod).

...comparisons to pre-injury benchmarks **...display residual deficits in jump height, power and RSI-Mod.**

Of note, increases in performance compared to pre-injury values have been shown on the uninjured side, suggesting that comprehensive rehabilitation, including strength and power training, can maintain the capacity of the uninjured side.

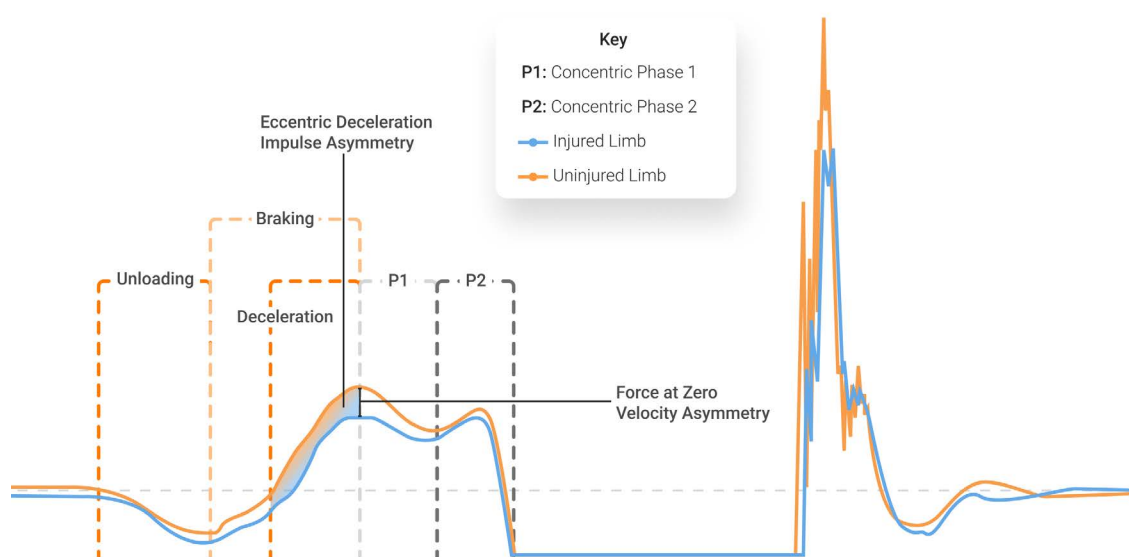
Bilateral Jumping Deficits Following ACLR

Kinetics

Decreased GRFs are a mechanical representation of reduced knee moments, shifting the demands away from the involved limb via an interlimb compensation. Most notably, between-limb deficits in eccentric deceleration and concentric impulse are observed.

Lower landing forces are also common (but often more variable). Cumulatively, these observations indicate compensatory strategies in load absorption and force production after ACLR.

...between-limb deficits in **eccentric deceleration and concentric impulse** are observed.



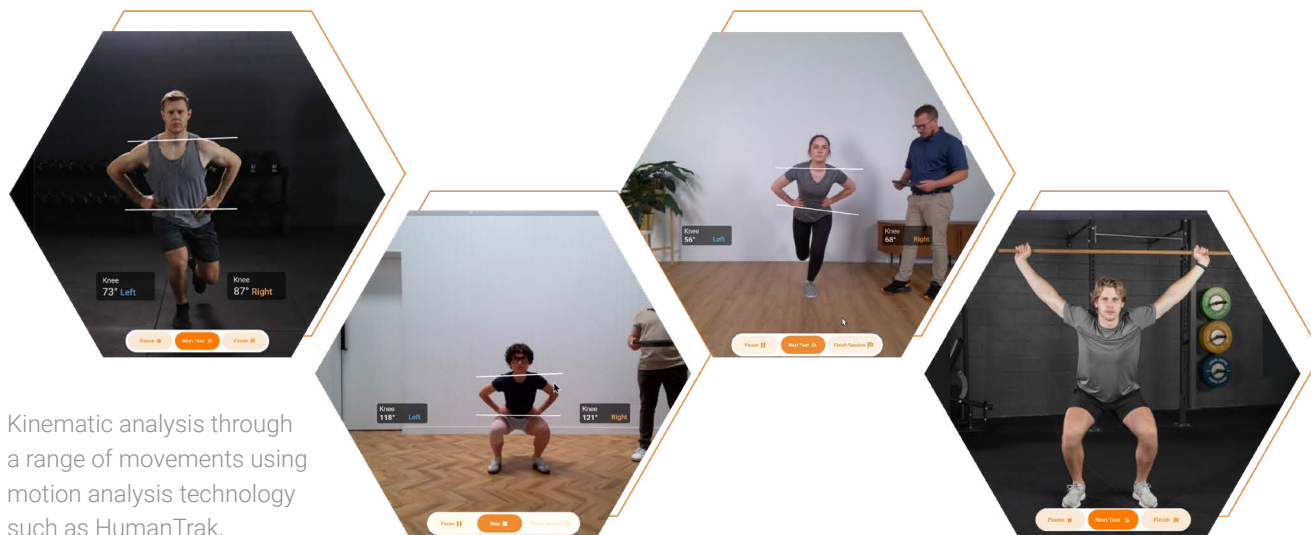
CMJ of an ACLR athlete showing deficits in various phases and metrics for the injured limb.

As time progresses post-surgery, the magnitude of asymmetry is typically reduced, but noticeable between-limb differences can remain beyond **nine months** or, in some cases, **several years post-surgery**. Athletes also appear to increase the braking phase duration and display lower GRF on the involved limb during the landing phase.

Athletes appear to increase the braking phase duration and display lower GRF on the involved limb during the landing phase.

Kinematics

Intra-limb compensation strategies are also commonly observed after ACLR in which reduced function at the knee is compensated for by altering the kinematic strategy to allow for a higher proportion of work done at proximal or distal joints (e.g., hip or ankle). These asymmetries are evident in the sagittal plane.



Kinematic analysis through a range of movements using motion analysis technology such as HumanTrak.

Typically, lower knee flexion angles on the involved side are present with a corresponding [increase in hip and trunk flexion](#). Collectively, athletes tend to adopt a landing strategy on the involved side with less knee ROM during landing after ACLR.

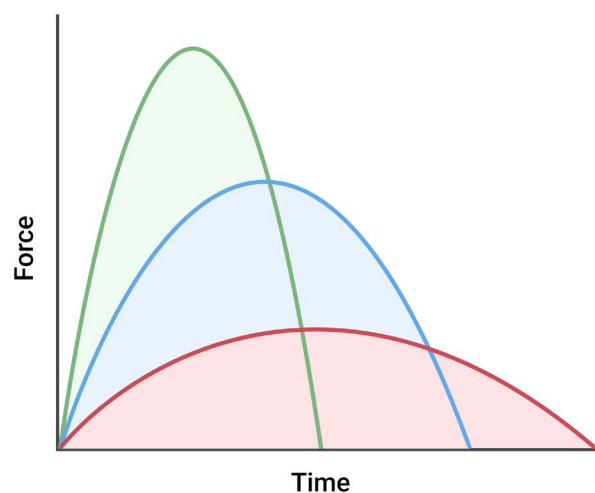
Understanding Movement Strategy

Jump height is an important metric and should be monitored during rehabilitation, but much like hop distance, the presence of residual deficits can be masked.

An assessment of function should examine the way in which the task was performed to provide a clearer indication of their readiness to re-perform.

While three different strategies may produce the same total amount of concentric impulse (and therefore jump height), there may be clear differences in how that impulse is generated.

This further emphasizes the need for in-depth strategy analysis.



Single Leg Performance Capacity

Single leg tasks, such as the SLJ, are more representative of limb capacity due to the greater relative force demands, often revealing larger residual deficits at the time of return to sport compared to [pre-injury values](#).

In contrast, the athlete is able to move at higher velocities during a bilateral CMJ, allowing practitioners to observe how athletes “offload” and adapt, particularly when compensating with the non-injured limb.

The lower relative force demands of the bilateral CMJ (during braking, takeoff and landing) also make it useful for assessing jump mechanics and identifying inter-limb compensations earlier in the rehabilitation process.

This information provides an objective measure of an athlete’s progress and helps guide future training prescriptions.

CMJ	Injured Athlete	Healthy Athlete
Concentric Velocity	2.7m/s	3.0m/s
Eccentric Velocity	1.0m/s	1.4m/s
Countermovement Depth	34cm	37cm
Concentric Duration	301ms	282ms
Eccentric Duration	642ms	526ms
% BW Load per Limb	68%BW	



SLJ	Injured Athlete		Healthy Athlete
	Injured Limb	Healthy Limb	
Concentric Velocity	2.06m/s	1.89m/s	2.15m/s
Eccentric Velocity	0.6m/s	0.73m/s	0.87m/s
Countermovement Depth	19.2cm	22.4cm	22.7cm
Concentric Duration	341ms	330ms	333ms
Eccentric Duration	603ms	592ms	526ms
% BW Load per Limb	84%BW		



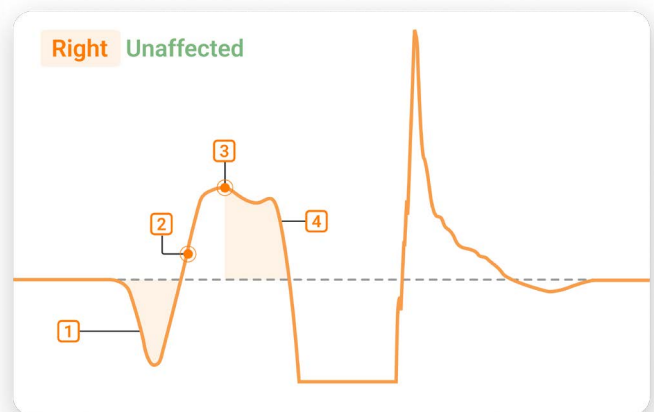
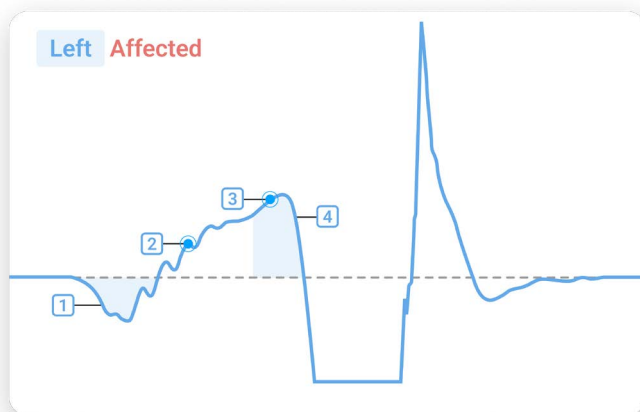
Differences in velocity, phase duration and countermovement depth between double leg and single leg CMJs in ACLR vs. healthy controls. (Modified from [Cohen et al., 2020](#)).

Although the CMJ will expose the athlete to greater velocities and, therefore, greater absolute deceleration demands, experience suggests that SLJs expose distinct deficits in an athlete’s ability to effectively decelerate. These unique demands present themselves in notable disruptions to the SLJ force-time curve following ACLR.

...SLJs expose distinct deficits in an athlete’s ability to effectively decelerate. **These unique demands present themselves in notable disruptions to the SLJ force-time curve following ACLR.**

This creates a cascade of events, beginning with lower force at the end of the eccentric phase (force at zero velocity), ineffective transfer between eccentric and concentric phases, and, thus, a lower jump height due to notable reductions in concentric impulse and subsequent takeoff velocity.

The SLJ Cascade



1 Inadequate Unweighting:

A shallow unweighting phase is common in athletes with poor deceleration capacity, as large unweighting impulses require an equal amount of deceleration impulse to redirect momentum.

2 Small Braking Forces and RFD:

After the minimal unweighting, the injured limb demonstrates poor neuromuscular function in its shaky braking phase, demonstrating an inability to decelerate momentum effectively.

3 Eccentric to Concentric Force Transfer:

Low eccentric peak forces lead to an over-reliance on concentric impulse to improve jump performance.

4 Jump Outcomes:

Lower amounts of net concentric impulse will lead to lower jump heights.

1 Rapid Unweighting:

A quick and large unweighting impulse allows an athlete to achieve a fast eccentric velocity, setting themselves up for high braking and deceleration RFD requirements.

2 Large Braking Forces and RFD:

After the rapid unweighting, an elite athlete is able to decelerate their momentum rapidly by producing large eccentric forces in short time periods.

3 Eccentric to Concentric Force Transfer:

High eccentric peak forces lead to effective concentric propulsion forces delivered from the SSC.

4 Jump Outcomes:

Greater amounts of net concentric impulse will lead to higher jump heights.

A typical SLJ force time curve from the involved (blue) and uninvolved (orange) limb of an athlete during the mid-stage of their rehabilitation following ACLR.

Factors Underpinning **Movement Deficits**

Strength deficits are common (especially in knee extension) following ACLR and can remain long after surgery. As previously mentioned, [research](#) shows that athletes with lower knee extension torque display the following:

- Reduced single leg drop jump performance (height and RSI-Mod).
- Shallower center of mass (CoM) displacement and peak knee flexion angles.
- Peak landing force occurring earlier during the ground contact phase compared to stronger athletes (see table below).

Metric	Significant difference between weak and strong athletes?
Jump Height	✓
Reactive Strength Index (RSI)	✓
Relative Mean Concentric Power	✓
Relative Mean Eccentric Power	✓
Peak CoM Displacement	✓
Concentric Impulse	✗
Eccentric Impulse	✗
Time of Peak Landing Force	✓
Force at Peak Displacement	✓

Single leg drop jump performance and mechanics in athletes of different strength levels. Adapted from [Maestroni et al. \(2022\)](#).

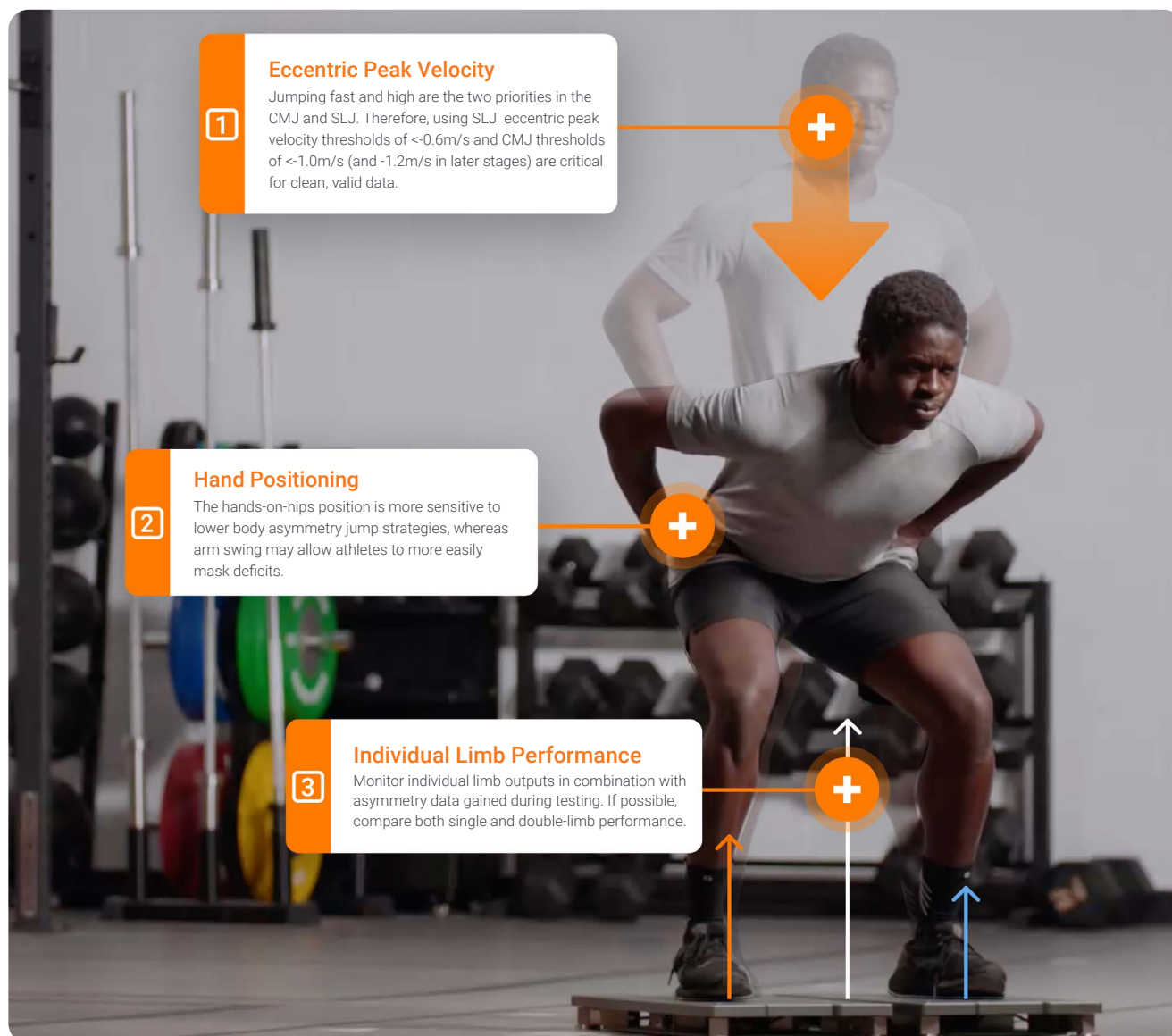
Shallow peak knee flexion angles during these tasks have been associated with increased injury risk, poorer tibiofemoral articular [cartilage composition and matrix degeneration following ACLR](#) and are characterized by an impaired capacity to effectively decelerate.

In the CMJ, stronger athletes often jump higher, display shorter braking durations, greater peak eccentric velocity and RSI-Mod than weaker players.

Therefore, strength measures of knee flexion and extension (e.g., measured by the DynaMo or ForceFrame) should be included to help characterize deficits in jump performance and targeted as part of a comprehensive rehabilitation program.

Considerations for Vertical Jump Tests

There are three main areas for consideration when performing and analyzing a vertical jump test:



Selecting Metrics That Matter

When assessing athletes post-ACLR, it is crucial to evaluate not just jump height but how the jump was performed. Practitioners can break metrics down into three key areas:

- **Output** (Performance) – The key result or outcome of an assessment (e.g., CMJ height), providing immediate feedback for an area of athlete focus.
- **Driver** – Mechanical factors (force and impulse) that influence performance and highlight areas for rehabilitation.
- **Strategy** – How the jump was executed, including phase duration, CoM displacement and velocity, offering context for the output and driver.

Asymmetry of driver variables can provide additional information regarding compensatory strategies and “offloading.” However, the reliability of these metrics should be monitored and measurement of the individual limb values should be the primary focus.

In single-leg jumps, it is also advised to assess peak landing force relative to jump height. Avoid using absolute peak force, as higher jumps on the non-involved limb can lead to greater impact velocity and GRF.

Right: Uninvolved				Left: Involved	
Metric	Output			Metric	Output
Jump Height	18cm			Jump Height	10cm
Peak Landing Force	2815N			Peak Landing Force	2512N
Landing Force Jump Height	156.3N/cm			Landing Force Jump Height	251.2N/cm

...assess peak landing force relative to jump height [in SLJs] ...as higher jumps on the non-involved limb **can lead to greater [GRFs].**

Does Surgery Type **Make a Difference?**

When assessing and rehabilitating athletes, it is important to understand the potential for differences in function in order to appropriately forecast the trajectory of recovery.

In a bilateral CMJ, elevated between-limb asymmetry has been observed in eccentric and concentric impulse by athletes with a bone-patellar tendon-bone (BPTB) graft nine months post-surgery compared to a semitendinosus graft (ST).

Similar findings were also observed by Jordan et al. (2022), with slower progression in those with BPTB more than 300 days after ACLR. Practitioners may wish to target these asymmetries during rehabilitation and consider the apparent differences when forecasting realistic timeframes for neuromuscular recovery.

Surgical graft types (allografts and autografts), as well as harvest locations (patellar, quadriceps and hamstring tendons) **will...influence the athlete's progression during ACLR recovery.**

Surgical graft types (allografts and autografts), as well as harvest locations (patellar, quadriceps and hamstring tendons), will therefore influence the athlete's progression during ACLR recovery. Practitioners may consider comparing cohorts by graft type to better understand functional recovery timelines during ACL rehabilitation.

Other Jumps to Assess

Limiting jump testing solely to the CMJ may not effectively measure different physical characteristics. The CMJ is an assessment of slow stretch-shortening cycle (SSC) movement (>250 milliseconds ground contact time) and provides an indication of mechanical muscle power and elastic strength.

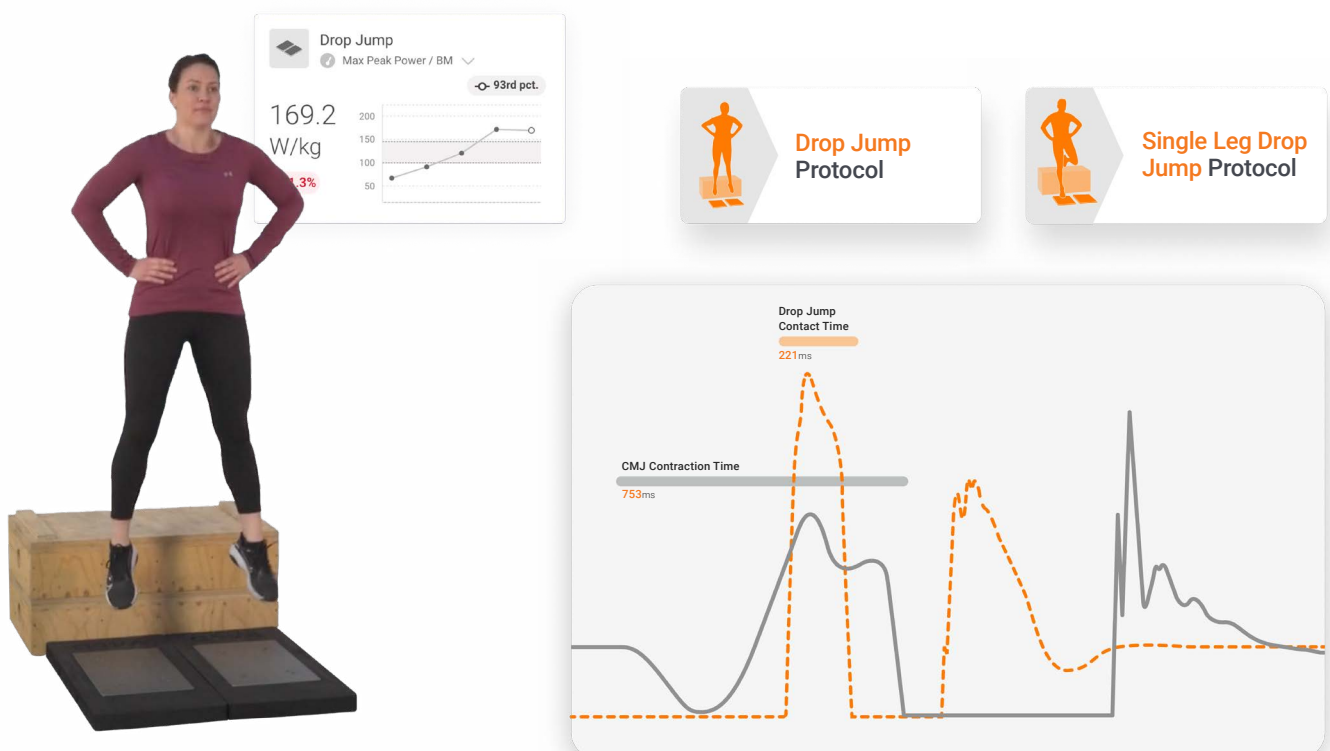
Including rebound tasks (e.g., drop jump) provides a measure of fast SSC (<250 milliseconds) function. These tasks are more related to reactive strength characteristics, in which the body must apply braking forces to decelerate its CoM prior to creating propulsive forces to move in a new direction, which has application for efficient COD and sprinting.

Including rebound tasks (e.g., drop jump) **provides a measure of fast SSC (<250 milliseconds) function.**

Drop Jumps

The drop jump is an assessment of fast SSC function and requires athletes to perform rapid transitions from eccentric to concentric actions, minimizing the period of amortization.

Reactive strength is impaired after ACLR in both double and single leg drop jumps, even following the completion of rehabilitation. Thus, identifying deficits in rebound capacity can be useful for informing reconditioning programs to optimize return to sport readiness.



Key Considerations for **Rebound Tasks**

What **Drop Height** Should I Use with My Athletes?

In a drop jump assessment, practitioners must consider what is the most appropriate drop height for their athletes. Intuition leads us to standardize the box height to 30cm for a bilateral test to allow for comparisons with previous research. However, the basis for this height is not well established, and when screening a range of athletes, different heights may provide either insufficient or excessive forces due to variations in neuromuscular qualities.



Test Consideration



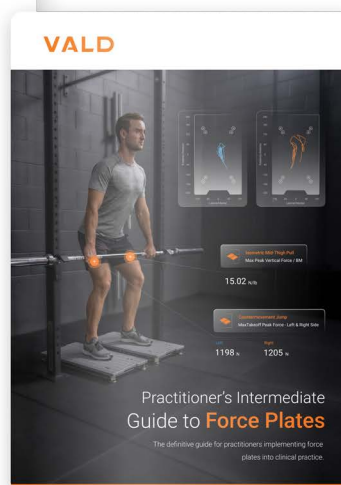
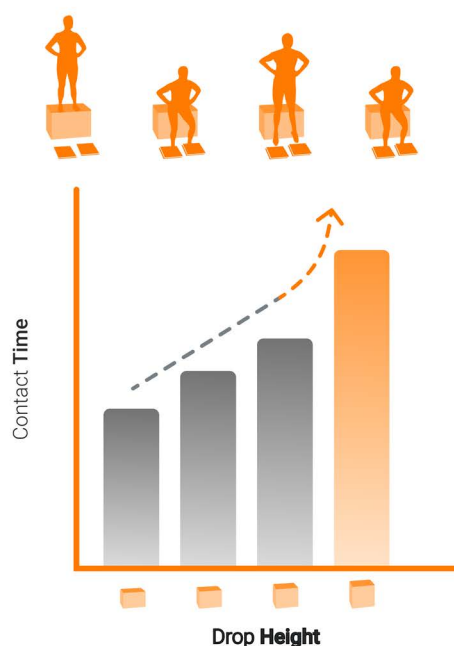
Make sure to factor force plate height into box height calculations. ForceDecks are ~5.5cm (2.2in) in height from the floor and should be subtracted from the box height the athlete is dropping from for accuracy.

The countermovement rebound jump (CMRJ) is a variation to overcome this constraint, providing a height reflective of their own neuromuscular ability (i.e., jump height) and a more perturbed landing position compared to a drop jump from a set height.

These assessments may better represent the ability of the neuromuscular system to provide force attenuation in response to an individual's jumping capabilities.

However, it should be noted that evidence is not currently available to examine the utility of this assessment in athletes following ACLR.

Nonetheless, the drop jump is a worthwhile assessment and permits greater standardization, allowing opportunities for longitudinal measurement of landing mechanics and propulsive actions, whereby changes (or not) can be more clearly examined due to the fixed drop height.



For more information on finding an optimal drop jump height, check out insights on drop jump profiling in the [Practitioner's Intermediate Guide to Force Plates](#).

Hop Test

A hop test (or continuous rebound or repeated hop test) can also be performed in which athletes complete vertical jumps in place repeatedly with the aim of minimizing the time spent on the ground while jumping as high as possible. Test duration can be based on a set number of jumps or on time.

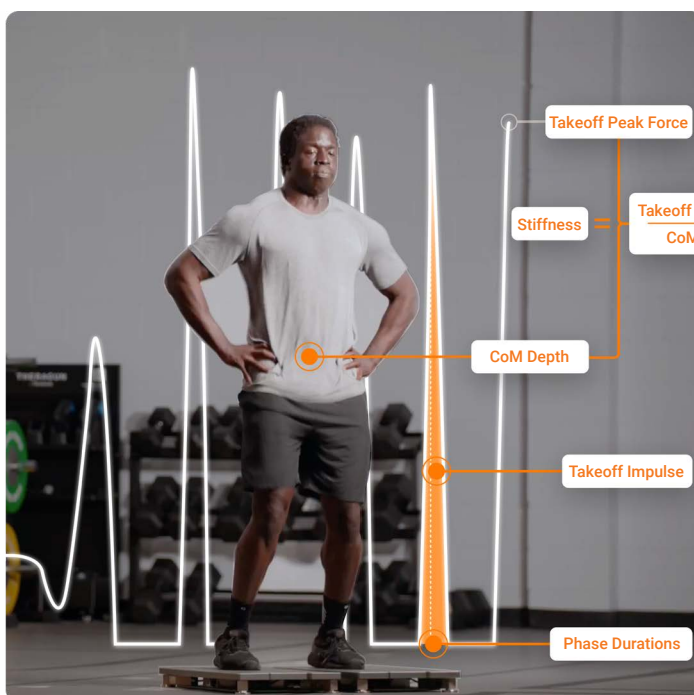
It is recommended to standardize the number of jumps to allow more accurate monitoring over time, and [previous research](#) indicates this can improve reliability. A single-leg version of this test has been used previously in male soccer players >6 months post-ACLR, displaying significant between-limb differences in jump height and RSI.

What Metrics Are Important to Measure in **Rebound Tests**?

RSI (jump height or flight time divided by ground contact time) is the metric that is most widely monitored during rebound tasks to identify residual deficits following ACLR and associations with key sporting actions such as COD. However, the values that make up the index should also be reported.

For example, the measurement of ground contact time ensures that the appropriate SSC actions are being assessed (e.g., <250ms for bilateral drop jumps to measure fast SSC). It should be noted that RSI is a ratio metric, and omitting the component parts does not permit an understanding of how the task was performed.

...measurement of ground contact time ensures that the **appropriate SSC actions are being assessed** (e.g., <250ms for bilateral drop jump to measure fast SSC).



RSI can improve across ACL rehabilitation phases, but several kinetic variables may not.

Therefore, it is also important to understand the movement strategy and mechanics via the inclusion of other variables listed.

Examine the **Force-Time Curve**

Analysis of the force-time curve provides rich information to understand the mechanics of the jump.

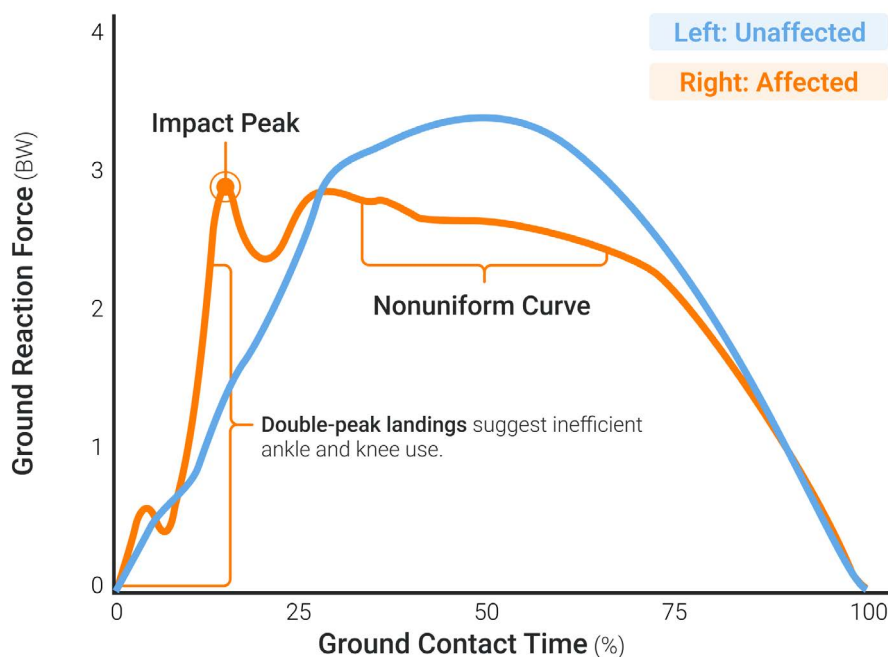
Post-ACLR, when athletes perform single leg drop jumps on their involved side, they are often categorized as **poor jumpers** (presence of an impact peak in the first 20% of the ground contact phase and a flatter curve).

...poor jumpers [display] an impact peak in the **first 20% of the ground contact phase and a flatter curve.**

More effective SSC mechanics are observed in the uninjured limb and in healthy controls, who are often classified as good jumpers. These mechanics are characterized by the absence of an impact peak (peak force occurring in the first 20% of ground contact) and an inverted U-shaped force-time curve, where GRF mirrors the CoM displacement, with peak force occurring around 50% of ground contact.

This is shown visually in the single leg drop jump force-time curve below. Underpinning factors appear to include reductions in quad and calf strength, and less efficient stretch shortening cycle mechanics.

A “poor” jump strategy may also be associated with an increased risk of reinjury, but further research is needed to examine this in more detail. As you become more comfortable assessing specific metrics, it is recommended to visually inspect the force-time curve to provide hints into other metrics that may warrant investigation.



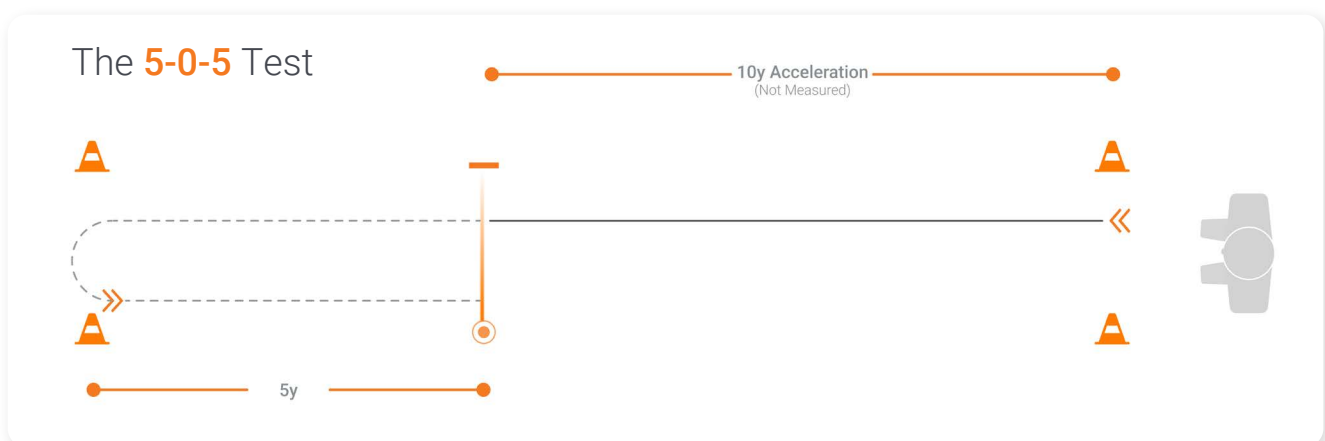
Single leg drop jump force-time curves on the involved (orange) and uninjured (blue) limb, indicating a poor and good jump landing classification respectively. Note the impact peak in the first 20% of ground contact on the involved side and absence of a U-shaped force-time curve.

Change of Direction Performance

COD is a key physical quality for success in a range of sports and is, in many cases, the mechanism of non-contact ACL injury. Field-based testing protocols that have commonly been used to assess [COD performance](#) during rehabilitation include [t-test](#), [Illinois agility](#) and [5-0-5](#).

However, the traditional application of these tasks does not isolate an athlete's ability to change direction effectively.

For example, acceleration is also examined, and as the duration of the test increases, there is a greater emphasis on anaerobic capacity and linear sprinting. Only 31% of the time spent performing a 5-0-5 test (involving a 180° action) is used to execute the [COD component](#).



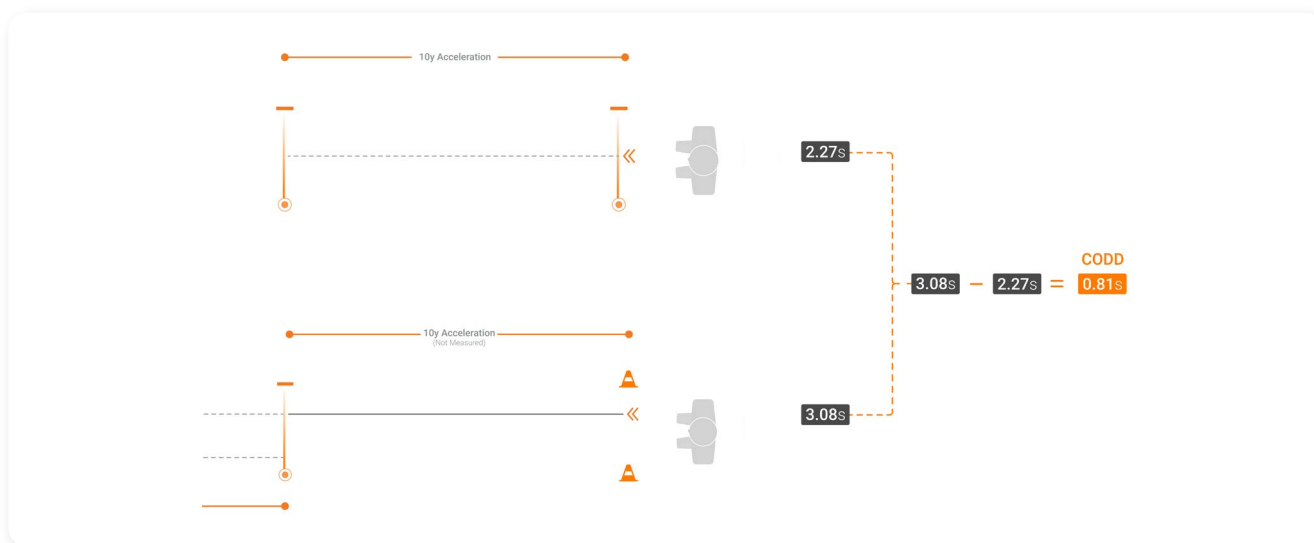
As COD speed is comprised of many constituent parts, including linear speed, deceleration and reacceleration, it is important to isolate these components as an athlete may have varying capabilities in each of these. Providing greater insights into how the performance task is executed will enhance the ability to identify the main limiting factor.

...COD speed is comprised of many constituent parts, **including linear speed, deceleration and reacceleration**, it is important to isolate these components...

It is recommended to also test linear sprint speed for athletes returning to running-based sports. A test comprising 10 and 30 meters using [timing gates](#) such as [SmartSpeed](#) will provide a profile of acceleration and maximal velocity running performance.

The [COD deficit \(CDD\)](#) can help to determine if an athlete should focus on their linear speed capabilities or ability to decelerate, change direction and reaccelerate during the training process.

CODD is calculated by recording the 5-0-5 time minus the first 10-m time taken during a 15m linear sprint of the 5-0-5 or (it could also be from a separate linear 10m sprint). This is mainly in cases where the modified 5-0-5 is removed, as this protocol removes the approach run.



Athletes returning to sport can be benchmarked against their peers on the COD total time and CODD as a marker of readiness for transition through phases of on-field rehabilitation and return to competitive training.

Female	10y Sprint	5-0-5		CODD	
		Right	Left	Right	Left
Cricket	2.06s	2.65s	2.65s	0.59s	0.59s
Netball	1.97s	2.50s	2.52s	0.53s	0.55s
Soccer	2.14s	2.67s	2.67s	0.53s	0.53s

Male	10y Sprint	5-0-5		CODD	
		Right	Left	Right	Left
Basketball	1.85s	2.43s	2.49s	0.58s	0.64s
Cricket	1.88s	2.4s	2.46s	0.52s	0.58s
Soccer	1.93s	2.4s	2.42s	0.47s	0.49s

Normative data from active athletes in soccer, cricket, netball and basketball for key speed related assessments (Dos'Santos et al., 2018).

COD Performance vs. Strategy

Time to complete the task has often been the [sole marker](#) for COD assessment during rehabilitation after ACLR. Research has shown biomechanical deficits when comparing the involved vs. uninvolved limb during a 90° cutting task despite [no differences in performance](#) time.

We recommend that while COD time and CODD assessed using timing gates should be included as markers of performance, other aspects should be considered. This includes body positioning and kinematics during a COD movement, deceleration requirements, the COD angle, running speed and reactive versus planned movements.

It is important to include qualitatively assessed kinematic variables (e.g., trunk position, foot placement, CoM height, knee angles, arm actions and visual focus) while executing the COD movement. A practically viable and validated solution is to include the [cutting movement assessment score \(CMAS\)](#).

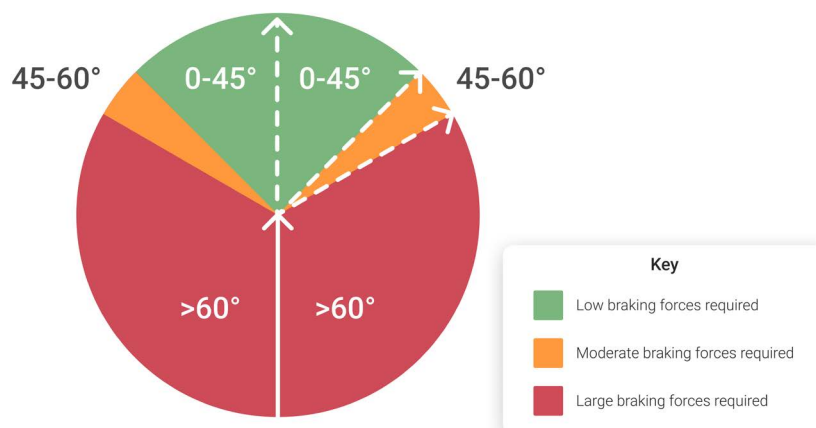
Considerations for COD Assessment

Importantly, the magnitude of the load placed on the knee joint will be largely determined by the COD angle required to perform the test. Higher cutting angles (i.e., $\geq 60^\circ$) require greater reductions in velocity to change the athletes' state of momentum.

This increases the braking forces, and thus athletes should be cleared for these assessments prior to gradual exposure to progressive deceleration and COD tasks from lower approach velocities and shallower cutting angles.

[Research](#) shows that reactive strength performance in assessments, such as a drop jump, is correlated with deceleration abilities. This allows practitioners to track metrics like RSI and concentric mean force during drop jumps to better inform readiness for COD tasks.

...performance [of] a drop jump, is correlated with deceleration abilities.
[Tracking] metrics like RSI and concentric mean force during drop jumps better informs readiness for COD tasks.

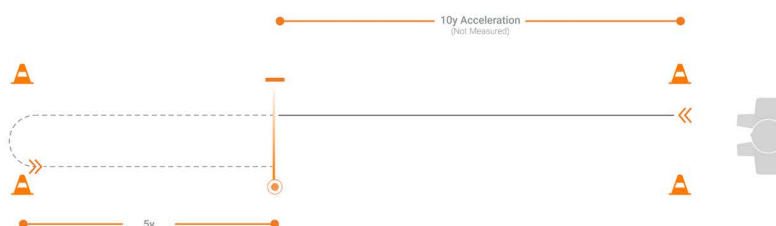


Higher cutting angles (i.e., $\geq 60^\circ$) require **greater reductions in velocity to change the athletes' state of momentum.**

A faster approach velocity will also increase the requirement to apply greater braking forces. The modified 5-0-5, in which athletes use a 5m approach prior to turning, may be used earlier than the traditional 5-0-5. The latter is characterized by a longer approach distance and allows them to increase their **entry velocity** (approximately 54% vs. 83% of max sprint speed in the modified vs. traditional 5-0-5, respectively).

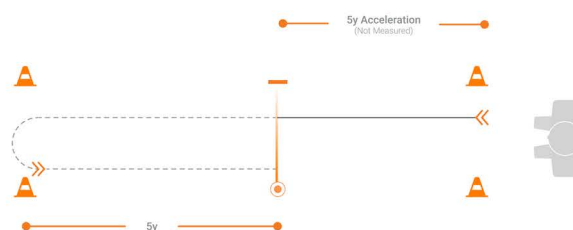
5-0-5

- 10y acceleration entry
- Faster entry speed
- Greater deceleration demands



Modified 5-0-5

- 5y acceleration entry
- Slower entry speed
- Lower deceleration demands



With an increased number of timing gates, approach velocity can also be measured, enabling the assessment of how athletes self-organize their approach speed and entry velocity based on their affordance control (how confident they feel to brake rapidly).

Allowing an athlete to self-select their approach will enhance the ecological validity and provide information as to their COD strategy. This can be monitored during rehabilitation to determine changes in performance and readiness to return to sport.

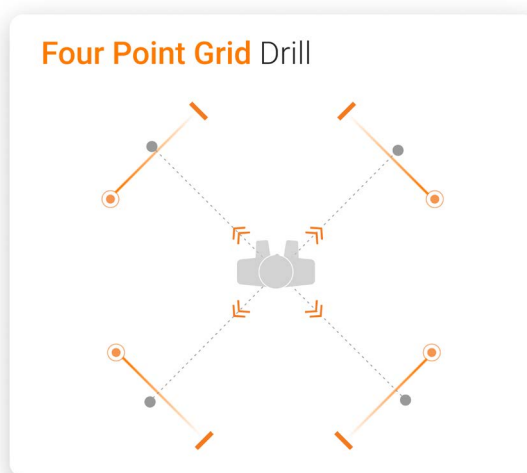
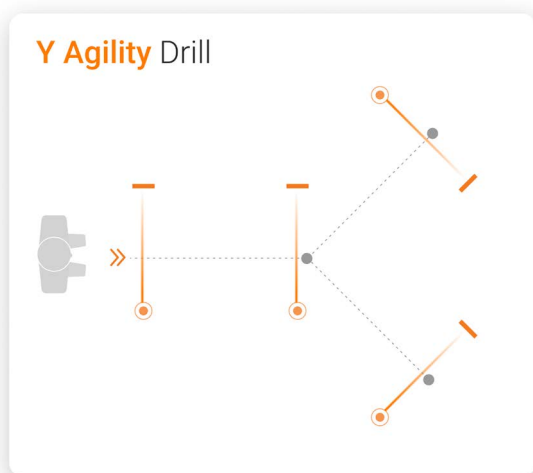
What is the difference between agility and COD, and should I measure it?

Non-contact ACL injuries often occur during direction changes, jump landings, or tasks like kicking, making COD speed measurements vital in ACL rehabilitation. COD speed is characterized by the ability to decelerate, reposition limbs and move in a new direction.

Non-contact ACL injuries often **occur during direction changes, jump landings, or tasks like kicking**, making COD speed measurements vital in ACL rehabilitation.

Agility has two components: speed in direction change and cognitive factors or a whole-body movement with change of velocity or direction in response to a stimulus.

Rarely in match play do rapid direction changes take place in optimal conditions where athletes have the luxury of time to select the appropriate movement strategy. Movement biomechanics are also likely to worsen during unplanned COD maneuvers due to the greater external load and mechanical stress placed on the knee joint.



Light-based or audio beep conditions are commonly used in unplanned task performance, offering some benefits, particularly in return-to-match scenarios where they create a training overload and simulate a worst-case scenario.

SmartSpeed, for example, can be effective in this context, providing athletes with a controlled environment to challenge their reaction times.

That said, these tests may have limitations when assessing sports-specific abilities. They don't fully replicate the dynamic nature of real sporting actions, where high-performance athletes rely on game knowledge and anticipate situations based on phase-of-play sequences.

As such, while they can be helpful, these tests may not capture the full integration of perception-action coupling and decision-making needed for effective COD tasks.

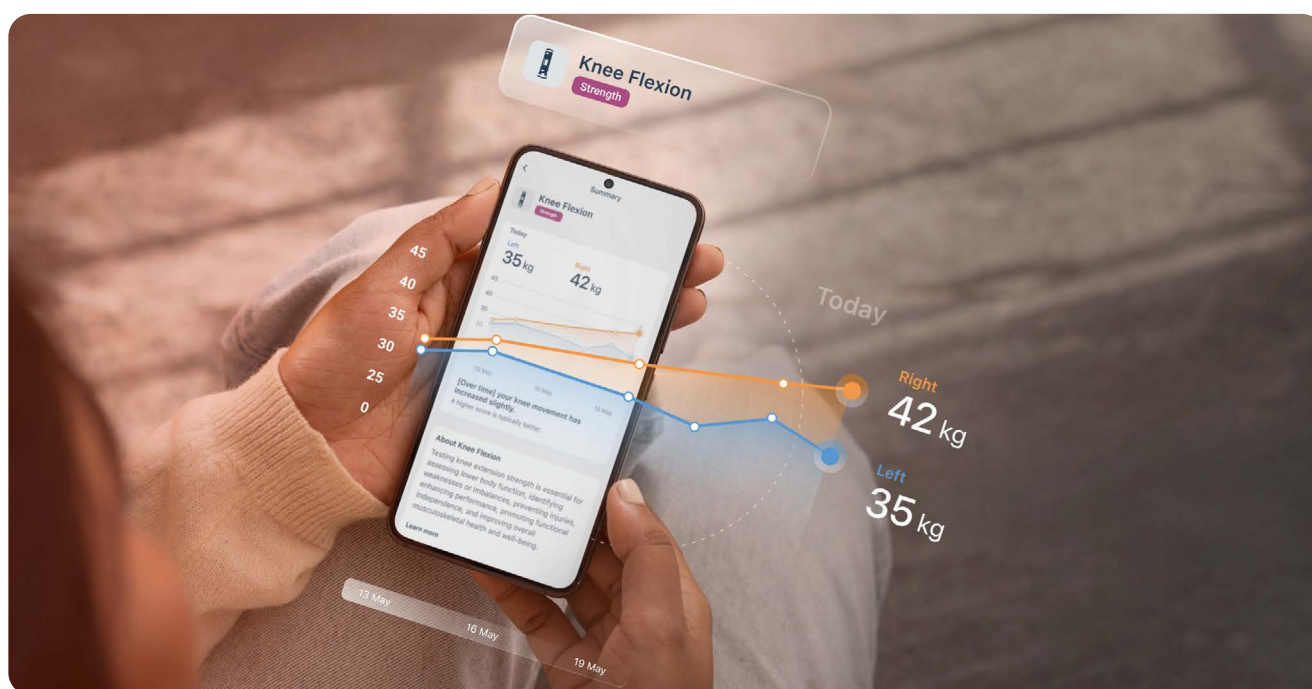
Building the Roadmap

A common approach to monitor progress during rehabilitation involves conducting assessments at standardized time points.

This has advantages in organizations with limitations in the number of treatment or assessment sessions athletes can access. The trajectory of recovery can also be compared between different groups considering surgical factors (graft type, secondary procedures) and decision modifiers (previous training age, level of competition, etc.).

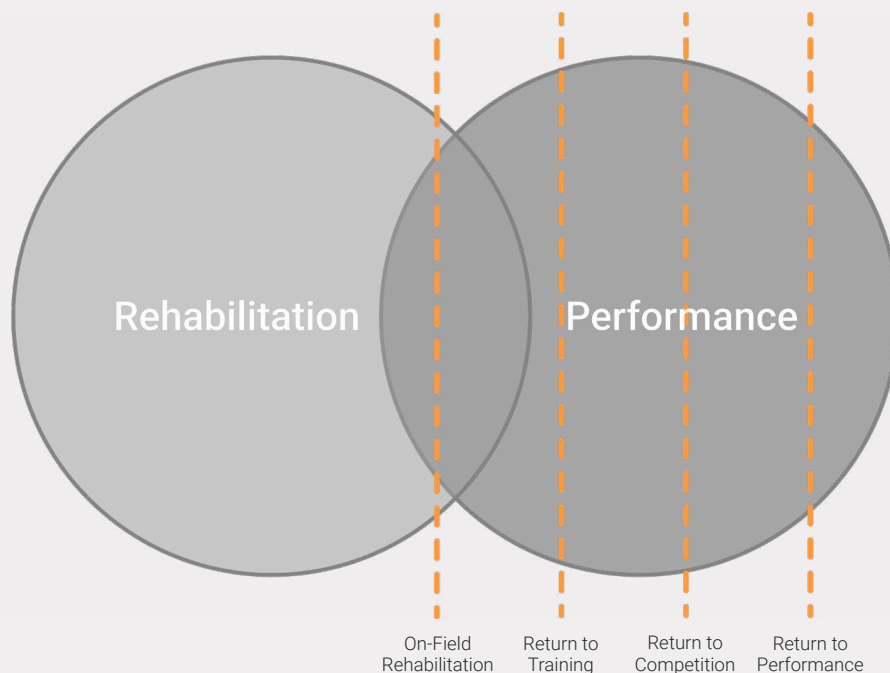
Information collected can be used to help set realistic benchmarks and design future treatment and training plans. When selecting time points for assessment, it is advised to allow sufficient periods between each to ensure meaningful physiological adaptations can occur that are outside of the normal variability in the test measures.

Information collected can be used to help set **realistic benchmarks and design future treatment and training plans.**



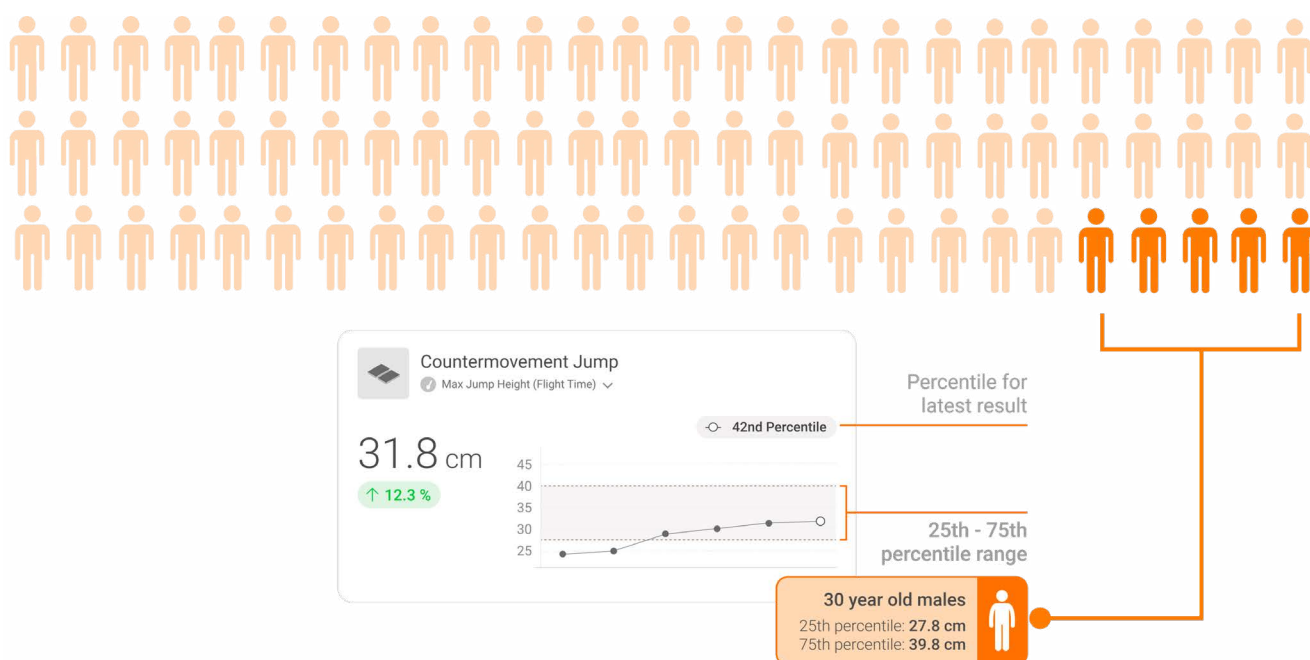
In clinical practice, progression through rehabilitation phases using a criterion-based approach is regarded as best practice.

Including clear criteria to progress through specific phases is beneficial to reduce the magnitude of deficits later in the athlete's journey, enhancing their readiness to re-perform. Starting early and regularly monitoring during rehabilitation can augment decision-making about when to progress to the next phase. Athletes can track their improvement and ensure a comprehensive recovery that addresses their specific deficits.



The phases of rehabilitation and return to sport, adapted from [Buckthorpe et al. \(2019\)](#).

However, it should be noted that evidence-based criteria to guide successful rehabilitation and return to sport following ACLR remain unclear. Addressing this knowledge gap is important, as consistent deficits have been identified during [return-to-sport testing](#). In their absence, it is prudent to use normative data to set benchmarks to guide phase progression.





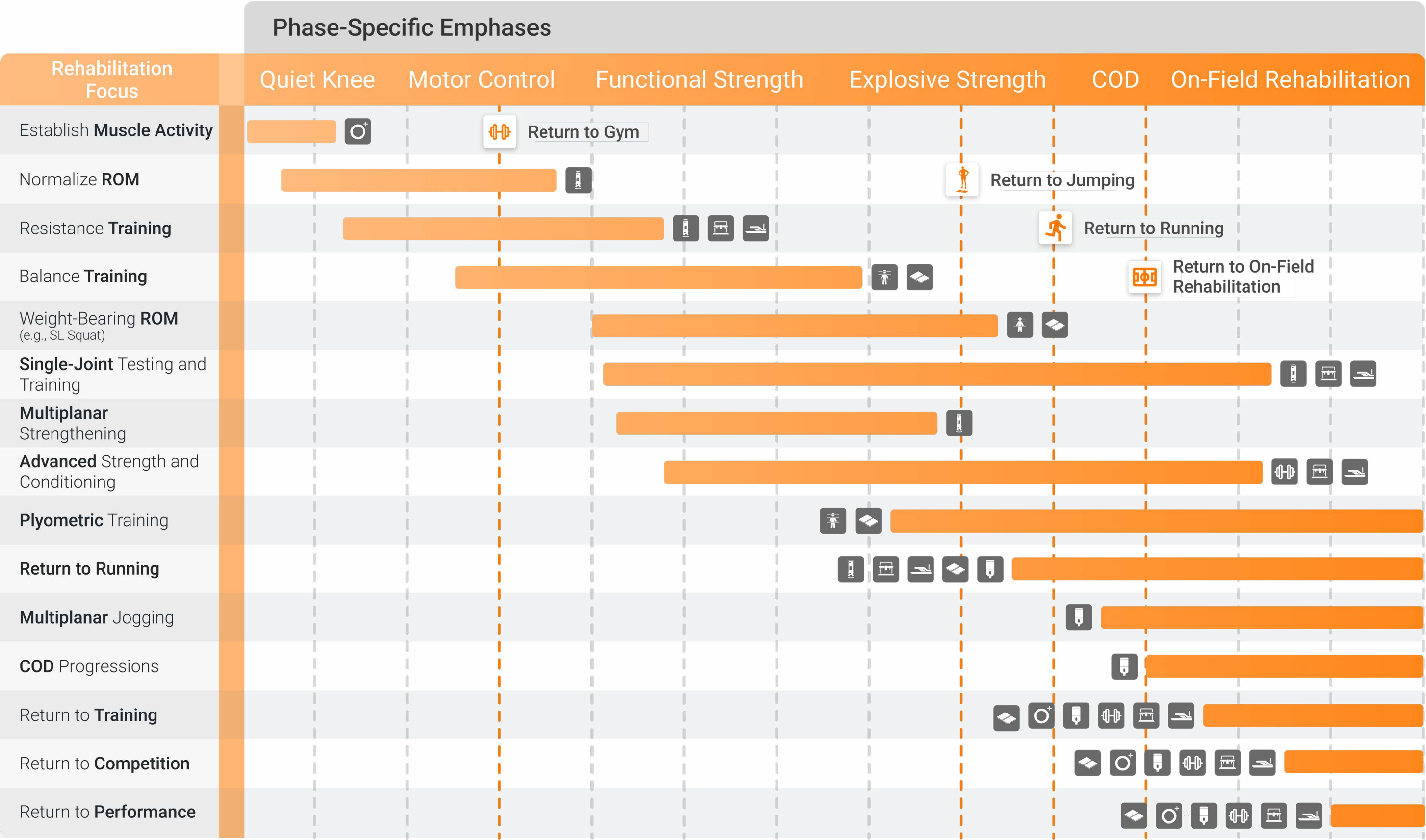
ACL Phase Progression

Paul Read, PhD, ASCC, CSCS*D, PGCE

Phase-specific emphases and measurement options from early rehab to return to performance during ACL rehabilitation.

Notes (pages 58-61)

Data are based on published (Read et al., 2019; Read et al., 2022; Kotsifaki et al., 2023) and unpublished research, as well as practical experience. Ranges reflect 95% confidence intervals. Timeframes may vary, but criteria-based progression should prioritize protection and tissue healing.





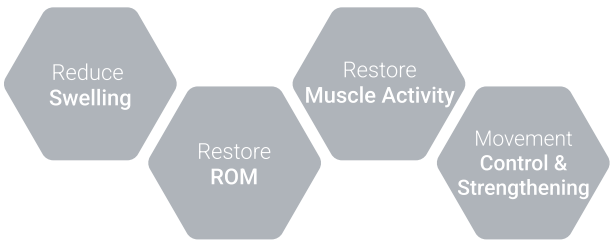
ACL Phase Benchmarks

Paul Read, PhD, ASCC, CSCS*D, PGCE

Criteria for phase-by-phase progressions in post-op ACL rehab, based on research and Paul Read’s expertise. For guidance only.

Early Phase (0-3 Months)

Rehabilitation Goals



Rehabilitation Milestones

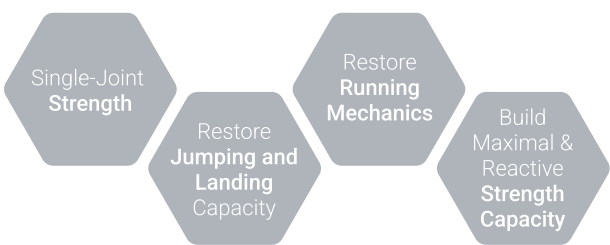
Note
Some gym work can begin earlier (posterior chain, hip, calf and trunk and targeting the uninvolved limb via cross training).

Return to Gym

Test	Benchmark
Swelling	None / minimal
Knee ROM	Extension <0°; flexion >130°
Straight Leg Raise	No extension lag
Squat Asym	<10% eccentric deceleration impulse asymmetry
SL Balance	<550mm total excursion
Glute Bridge	10 reps
Calf Raise	10 reps
Single Leg Squat	>70° knee flexion
Plantar Flexion Strength	>13N/kg
Knee Extension	>5.5N/kg
Knee Flexion	>2.0N/kg

Middle Phase (3-5 Months)

Rehabilitation Goals



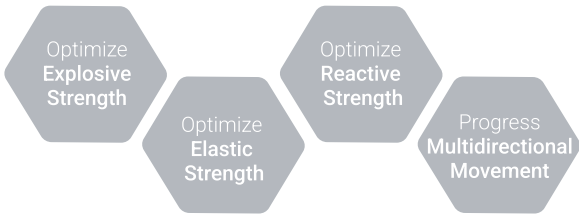
Rehabilitation Milestones

Return to Running and Jumping

Test	Benchmark
SL Calf Raise	>25 reps
SL Glute Bridge	>25 reps
SL Sit to Stand	15–25 reps
Single Leg Squat	>90° knee flexion
CMJ	27.8–30.1cm jump height; 9.2–13.4% concentric impulse asymmetry; <-1.0m/s eccentric peak velocity
SLJ	>11cm jump height; <20% concentric impulse asymmetry; <-0.5m/s eccentric peak velocity
Hop Test	<10% impulse asymmetry
Plantar Flexion	>15.5N/kg
Knee Extension	>7.0N/kg
Knee Flexion	>2.5N/kg

Late Phase (5-8 Months)

Rehabilitation Goals



Rehabilitation Milestones

Return to On-Field Rehabilitation

Test	Benchmark
CMJ	30.7–33.4cm jump height >2.25Ns/kg net concentric impulse / BM >1.35Ns/kg net eccentric deceleration impulse / BM
SLJ	>13.2cm jump height >1.5Ns/kg net concentric impulse / BM >0.6Ns/kg net eccentric deceleration impulse / BM
Drop Jump	>28cm jump height <10% eccentric and concentric impulse asymmetry
Plantar Flexion	>18N/kg
Knee Extension	>7.5N/kg
Knee Flexion	>2.8N/kg
IMTP	>32N/kg

Return to Train Phase (8-9 Months)

Rehabilitation Goals



Rehabilitation Milestones

Return to Training

Test	Benchmark
CMJ	31.8–34.7cm jump height >2.4Ns/kg net concentric impulse / BM >1.5Ns/kg net eccentric deceleration impulse / BM
SLJ	>15cm jump height >1.58Ns/kg net concentric impulse / BM >0.68Ns/kg net eccentric deceleration impulse / BM
Single Leg Drop Jump	>14cm jump height >0.38 RSI
IMTP	>34N/kg
Field Tests	COD CMAS <3 (no high-risk cutting maneuvers), 5-0-5 CODD, 10y and 30y sprint assessments >90% pre-injury levels

Return to Competition Phase (>9 Months*)

Rehabilitation Goals



Rehabilitation Milestones

Return to Competition

Test	Benchmark
CMJ	33.9–35.0cm jump height >2.6Ns/kg net concentric impulse / BM >1.65Ns/kg net eccentric deceleration impulse / BM
SLJ	>18cm jump height >1.8Ns/kg net concentric impulse / BM >0.73Ns/kg net eccentric deceleration impulse / BM
Single Leg Drop Jump	>17cm jump height >0.57 RSI
IMTP	>37N/kg
Field Tests	COD CMAS <3 (no high-risk cutting maneuvers), 5-0-5 CODD, 10y and 30y sprint assessments >95% pre-injury levels

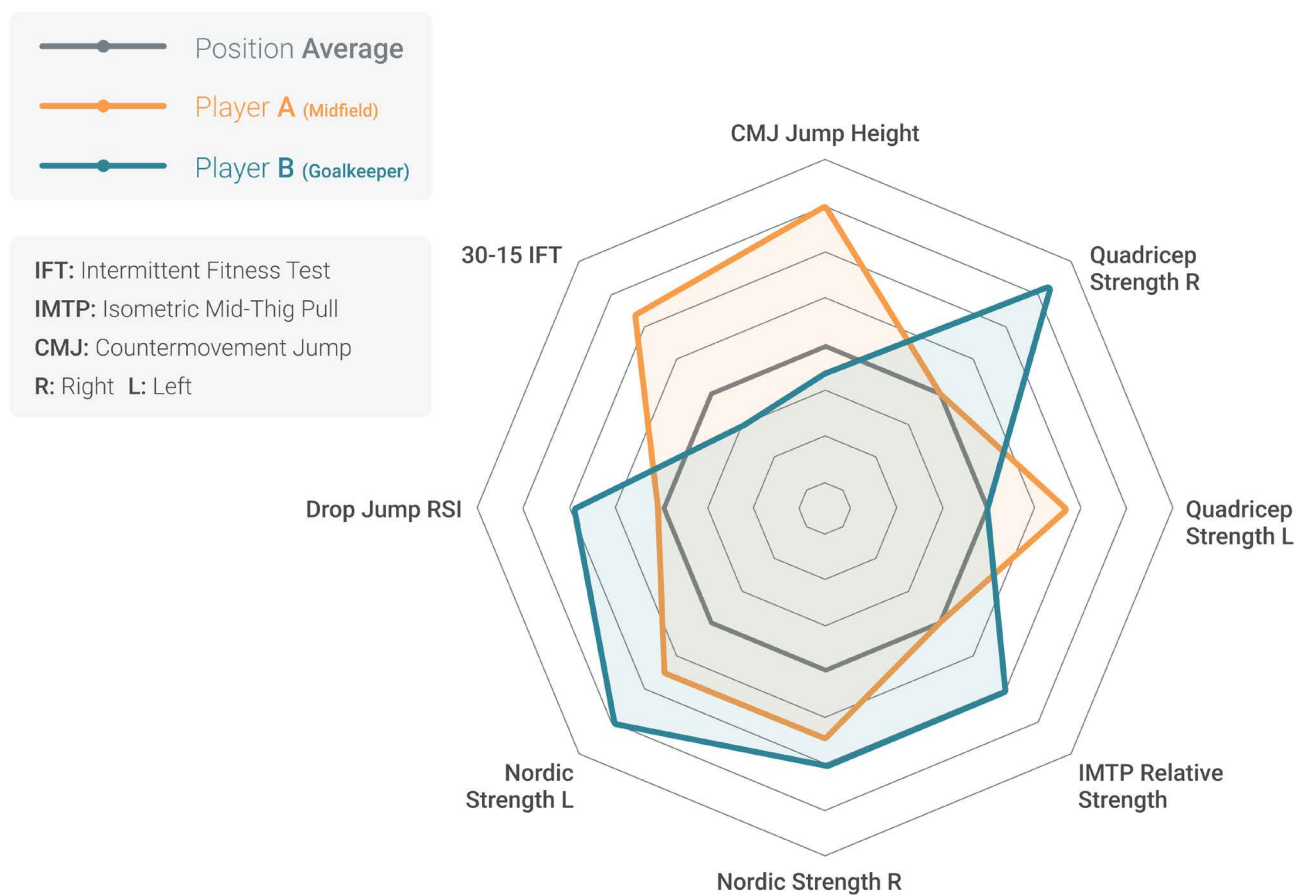
* Quality training and preventive care after formal ACL rehabilitation should emphasize the gradual development and maintenance of a relatively high chronic training load and implementing strategies to minimize the risk of reinjury through continued monitoring.

Enhancing Return-to-Sport Decision Making

Typical practice in return-to-sport testing involves passing a battery of tests (often hops) with the aim of achieving a 90% limb symmetry index (LSI).

This approach does not consider potential performance decrements on the uninvolved limb after an injury. Another concern is when athletes are asked to perform similar tests with a specific “pass” threshold, the probability of failing a test by chance alone increases with each test that is added to a battery.

Normalized Performance Profile



An alternative is to use a composite score encompassing different performance characteristics across a range of tests. This approach was originally used in fitness testing scenarios, using standardized (Z scores) from a series of tests to create a single total score of athleticism (TSA) value for each individual player.

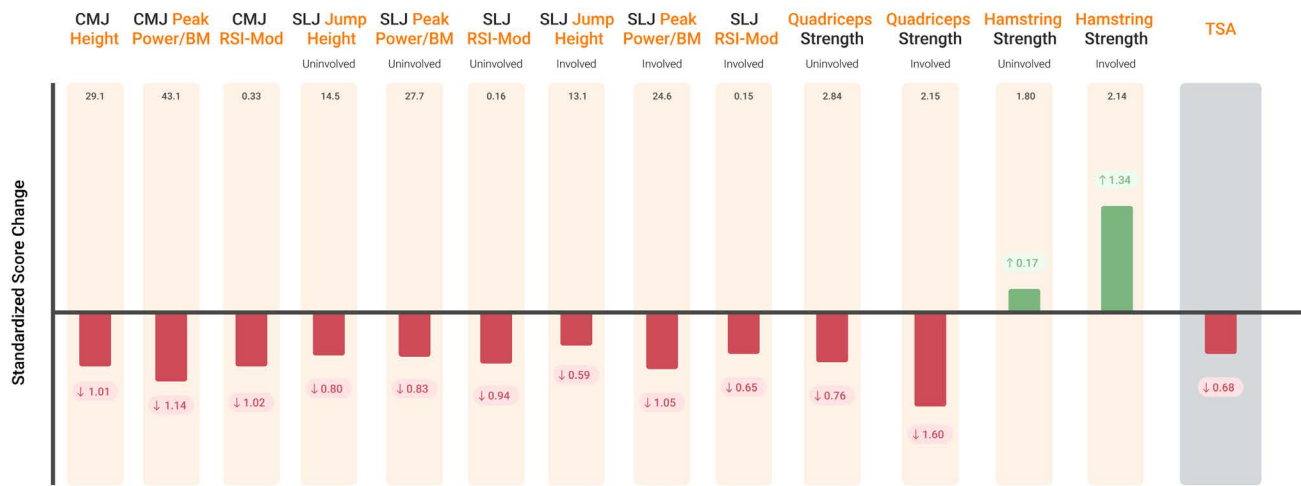
From this, clinicians and coaches can examine contextualized data of individual athletes relative to their teammates and set benchmarks for return-to-sport readiness that are realistic to the demands they will be exposed to.

The TSA has been used to profile athletes **returning to sport after ACLR**. Higher composite scores encompassing strength and power qualities like those discussed in this guide may be protective against further articular cartilage, meniscal and soft tissue injuries at the time of return to sport.

Measuring the TSA as part of return-to-sport profiling may, therefore, have a positive effect in lowering the risk of subsequent injuries and move beyond the limitations of limb symmetry assessment and generic benchmarks alone.



A. Athlete returning to sport with high physical capacity.



B. Athlete returning to sport with low physical capacity.

Figure from [Maestroni et al. \(2023\)](#). Athlete A displays the Z-scores for each test in a player with well-developed physical capacities and an above-average TSA score. Figure B represents a player with low physical performance and TSA score, indicating they are significantly below athletes at their level of competition and are therefore not ready to return to sport.

Case Studies



Matt Jordan
Jordan Strength

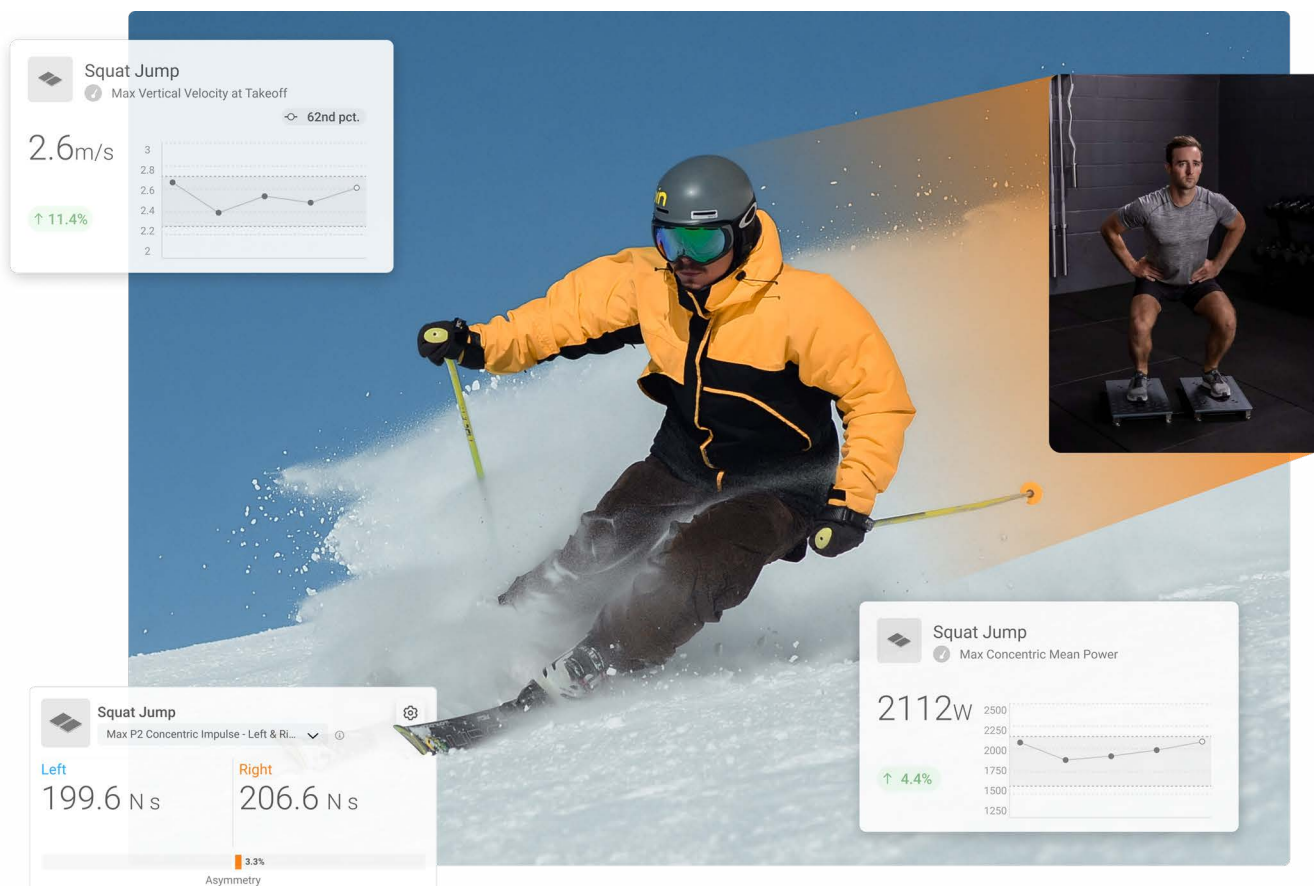
This novel, multidisciplinary [case study](#) includes several years of detailed analysis, including pre-injury and post-left ACLR neuromuscular testing, as well as training load data of an elite female alpine ski racer.

Approach to **Monitoring Functional** and **Neuromuscular Testing Outcomes**



A comprehensive suite of neuromuscular tests was used to track recovery milestones after ACLR. Specifically, we assessed force-velocity mechanics, CMJ function, SSC mechanics, rapid force capacity and performance fatigability using the [80s repeated SJ test](#).

These assessments can help practitioners assess an athlete's envelope of function after an ACL injury. The CMJs and SJs revealed improvements over time. By the return-to-sport phase, the eccentric deceleration impulse in the CMJ was comparable to baseline values, reflecting progress in the braking phase during SSC actions.

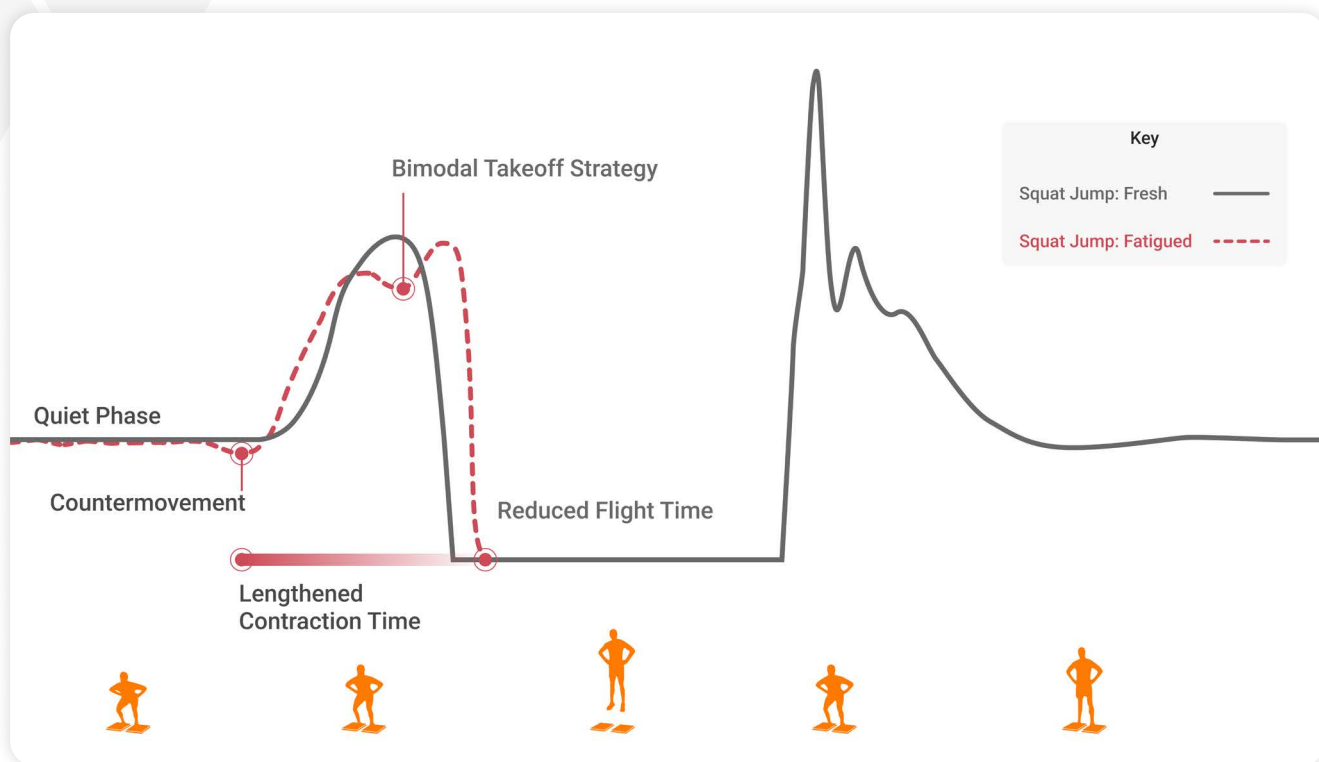


While asymmetries had normalized early in the return-to-sport transition, CMJ mechanical power and limb stiffness were reduced compared to the pre-injury period, even at 18 months post-surgery.

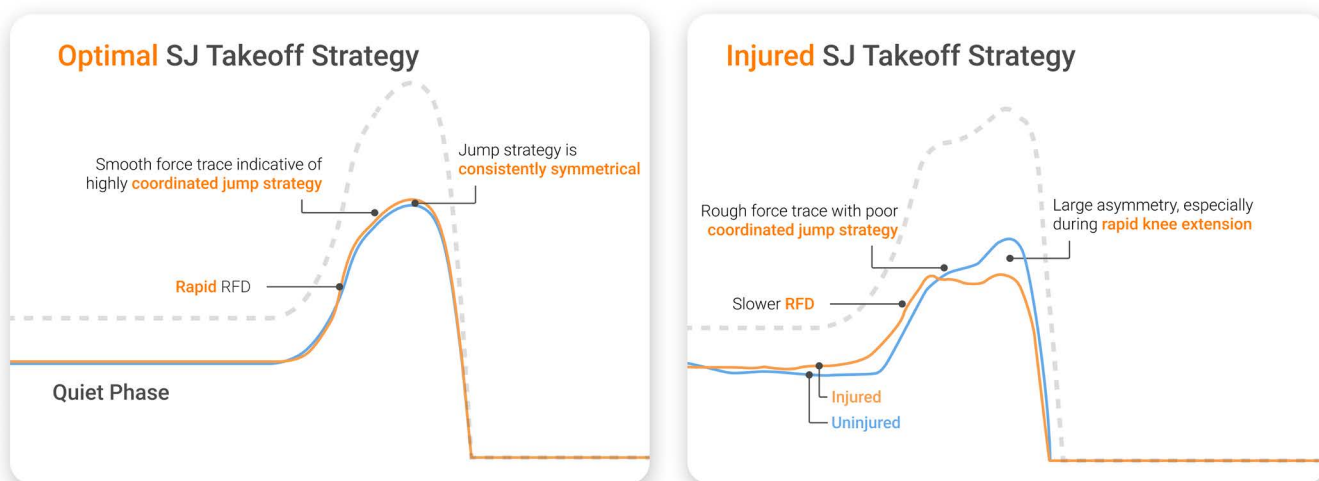
This raises questions about the practice of strictly using the contralateral limb benchmark to achieve a 90% LSI, as it fails to account for the absolute reduction in leg power output.

The 80s repeated SJ test was used to examine the effects of performance fatigability on power output. We developed this test to simulate a typical alpine ski race.

The test involves 20 maximal effort SJs that are preceded by a 4-second isometric hold in a 90° squat position.

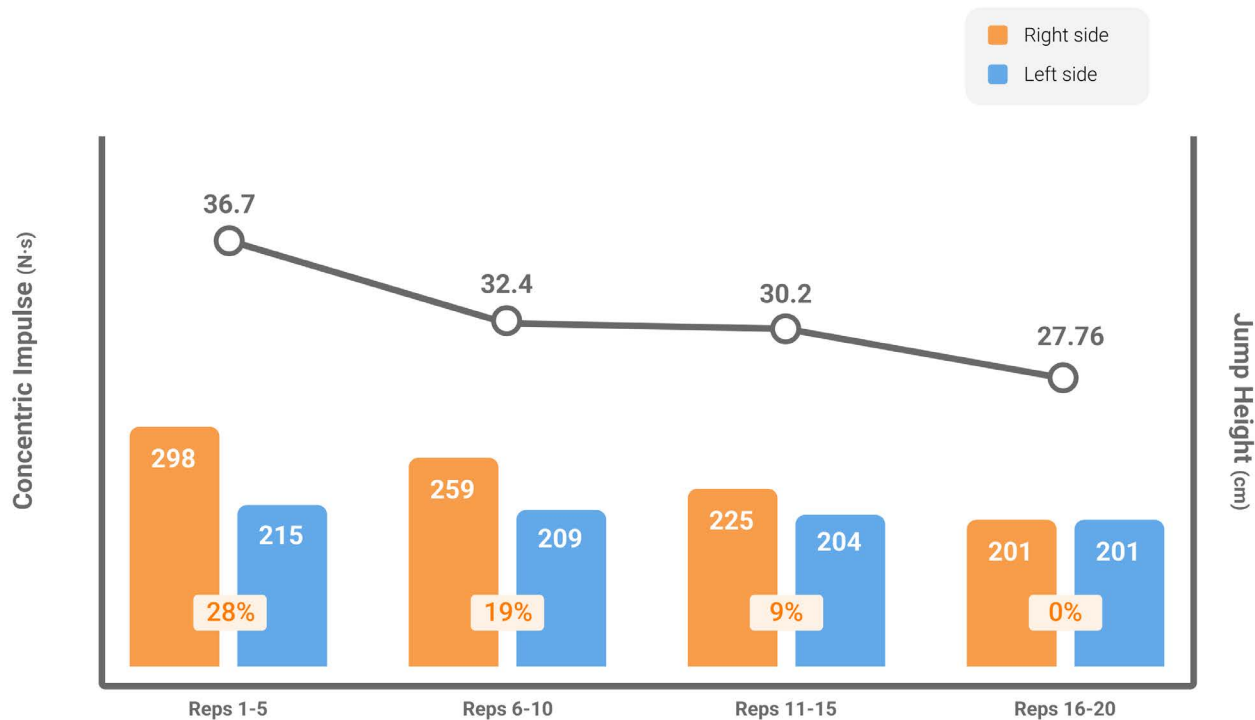


The test showed early fatigability in the contralateral limb and reduced force with a bimodal curve in the non-injured limb during the final SJ phase. This suggests the non-injured side compensated by generating more impulse to maintain vertical displacement.



This case underscores the importance of dynamic assessments that reflect real-world performance conditions, especially for sports like alpine skiing, where neuromuscular function under fatigue is critical.

This has implications for reinjury risk, especially for the uninjured side. The figure below shows how we represent the 80s repeated SJ test in our reporting.



Strength testing showed improved maximal strength in both ACLR and contralateral limbs, but the rate of torque development (RTD) remained below pre-injury levels, with little progress at 18 months.

The non-injured limb weakened over time, reducing torque and narrowing asymmetry but raising concerns about overall bilateral strength.



Contralateral declines in strength and power masked ACLR deficits, indicating incomplete recovery.

The results highlight the bilateral impact of ACL injuries, emphasizing the need to strengthen both limbs during rehabilitation to restore full function.



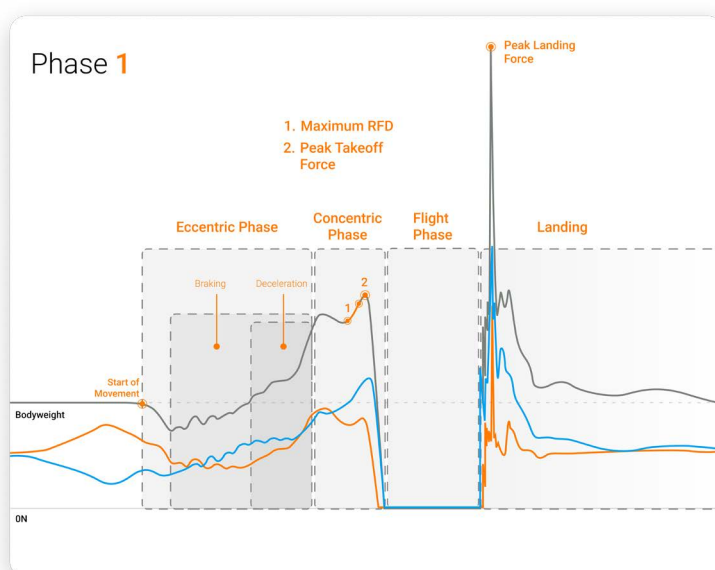
Travis Gaudet

ORKA Performance

A 20-year-old collegiate soccer player sustained a contact injury to the posterior knee during a league playoff game. After pre-operative baseline measures were assessed, the athlete decided to undergo ACL reconstructive surgery with a BPTB autograft and lateral meniscal repair.

This case addresses the **mid and later stages of the patient's ACL rehabilitation, after necessary timelines and criteria have been met, allowing for **safe dynamic and jump testing on force plate technology**.

Approach to **testing, monitoring** and **targeted exercise prescription**



Mid-Stage 1 (4 months)

After introducing CMJ and squat jump (SJ) assessments on ForceDecks, we analyzed the patient's physical capacities in three specific buckets to inform progressions, regressions and lateralizations for mid-stage exercise prescription.

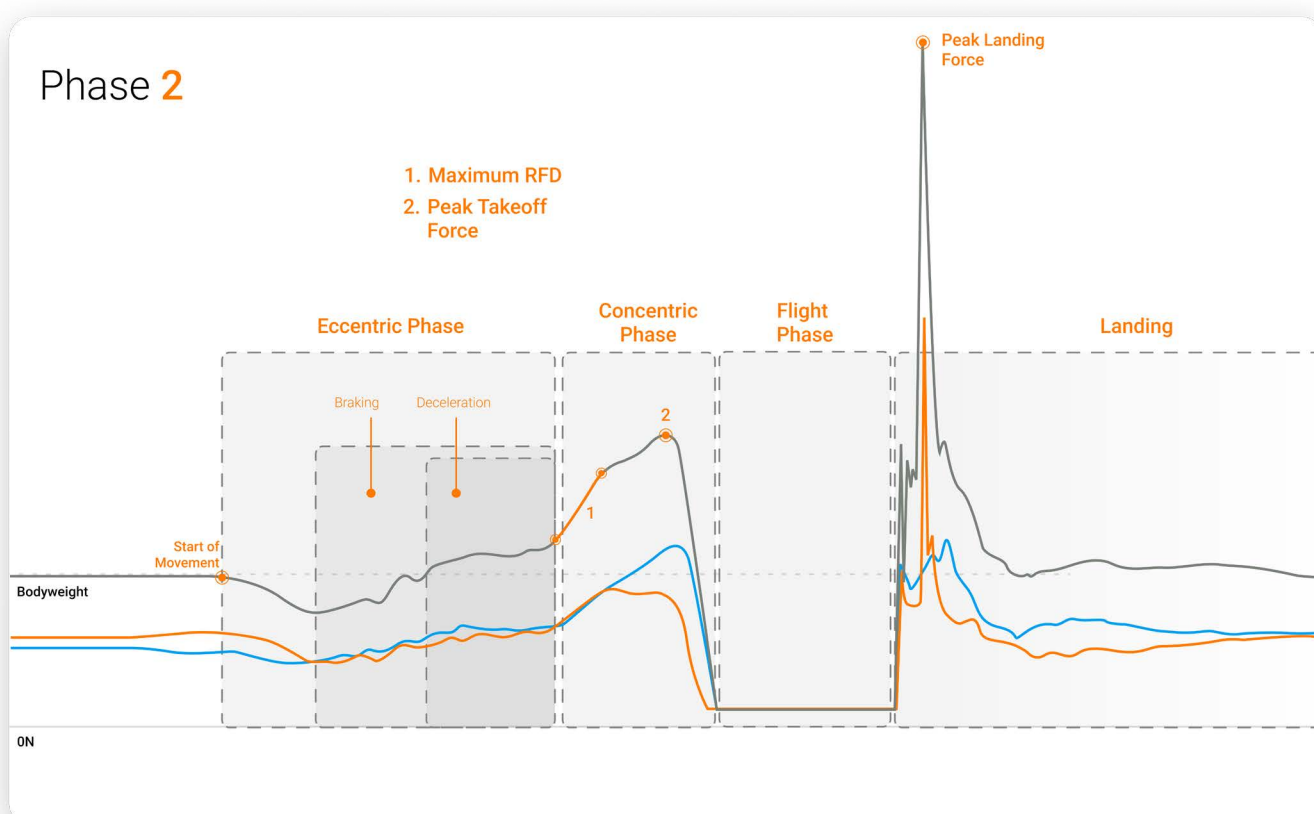
Addressing the Curve (Month 4) (examples of prescriptive stimuli)			
Quality	Eccentric Deceleration	Concentric Propulsion	Reactive Strength Capabilities
Intervention	Tall short	Explosive sit to stand	Double leg pogos (+/- banded assistance)
	Drop squat	SJ *then combining Eccentric and concentric portions into BW CMJ	Single leg pogos (+/- banded assistance)

The first bucket that needed addressing is eccentric deceleration. Poor unweighting, lack of countermovement depth, longer eccentric/concentric durations, low peak eccentric velocity (with normal targets ranging from 1.0–1.2 m/s) and poor eccentric deceleration/braking phases are common deficits seen in the mid-stages, frequently influencing challenges in jump performance.

Further in-depth assessments, including ROM, single-joint force production and neuromuscular coordination, can help determine the underlying variables clinicians can assess to intervene on rate-limiters to progression.

Mid-Stage 2 (5 months)

After 4 weeks of rehabilitation addressing the previously established buckets of need, CMJ reassessment was performed to determine progress in physical qualities.



Noting improvements in eccentric peak velocity, countermovement depth, contraction time and reductions in asymmetry during the CMJ, new exercise progressions were added to further enhance adaptations in the selected buckets of need.

Although asymmetries in the concentric phase of the CMJ are most apparent, the primary goal of addressing eccentric deceleration qualities continues to be addressed, guided by the initial reverse-engineering process.

Addressing the Curve (Month 4) (examples of prescriptive stimuli)			
Quality	Eccentric Deceleration	Concentric Propulsion	Reactive Strength Capabilities
Intervention	Trap bar deceleration	Weighted SJ/trap bar CMJ	Single leg pogos over band
	Bulgarian freeze drops	Bulgarian split squat jumps	Single leg pogos over cone

However, simply “progressing” may not be sufficient to completely inform clinician decision-making. Therefore, using large dataset comparisons to normative data, such as [VALD’s Norms](#), can help compare the athlete to their peers to best inform progressions and readiness for sporting activities.

Mid-Stage 3 (6-7 months)

With continued progression in CMJ physical characteristics, integration of fast SSC exercises, such as drop jumps and hopping, while continuing single leg progressions will help prepare the athlete for their return to sport training.

In these scenarios, typical underpinning limitations are identified through:

- A reduction in vertical foot/ankle stiffness and poor ankle dorsiflexion control.
- Inadequate strength-endurance (measured via single leg calf capacity for reps) or strength (seated calf test on ForceFrame or ForceDecks) and run-specific isometric tests (Run-Specific Ankle Iso-Push) relative to suggested benchmarks. (For more information on plantar flexor profiling, check out the [Practitioner’s Guide to the Calf and Achilles Complex](#)).

Following the intervention, progressions in force-time characteristics can be visualized based on the athlete’s changes in CMJ characteristics. Improvements in unweighting, eccentric deceleration impulse and RFD, force at zero velocity and concentric impulse can be visualized from the graphic below.

Addressing the Curve (Month 4) (examples of prescriptive stimuli)

Quality	Eccentric Deceleration	Concentric Propulsion	Reactive Strength Capabilities
Intervention	Banded double leg broad jump to single leg landing (overload)	Bulgarian split squat jumps in response to color/flashing arrow	Single leg tuck jump
	Weighted altitude drops with stroboscopic goggles (visuomotor disruption)	Double leg CMJ to single leg landing with therapist perturbation	Drop jump variations

Late Stage **Return-to-Play Progressions**

The athlete successfully re-integrated into sport-specific training, transitioning from linear movements (running) to multi-planar skills COD.

This approach included progressing through the control-chaos continuum while also attending technical training (restricted) under the guidance of the rehabilitation and performance team.

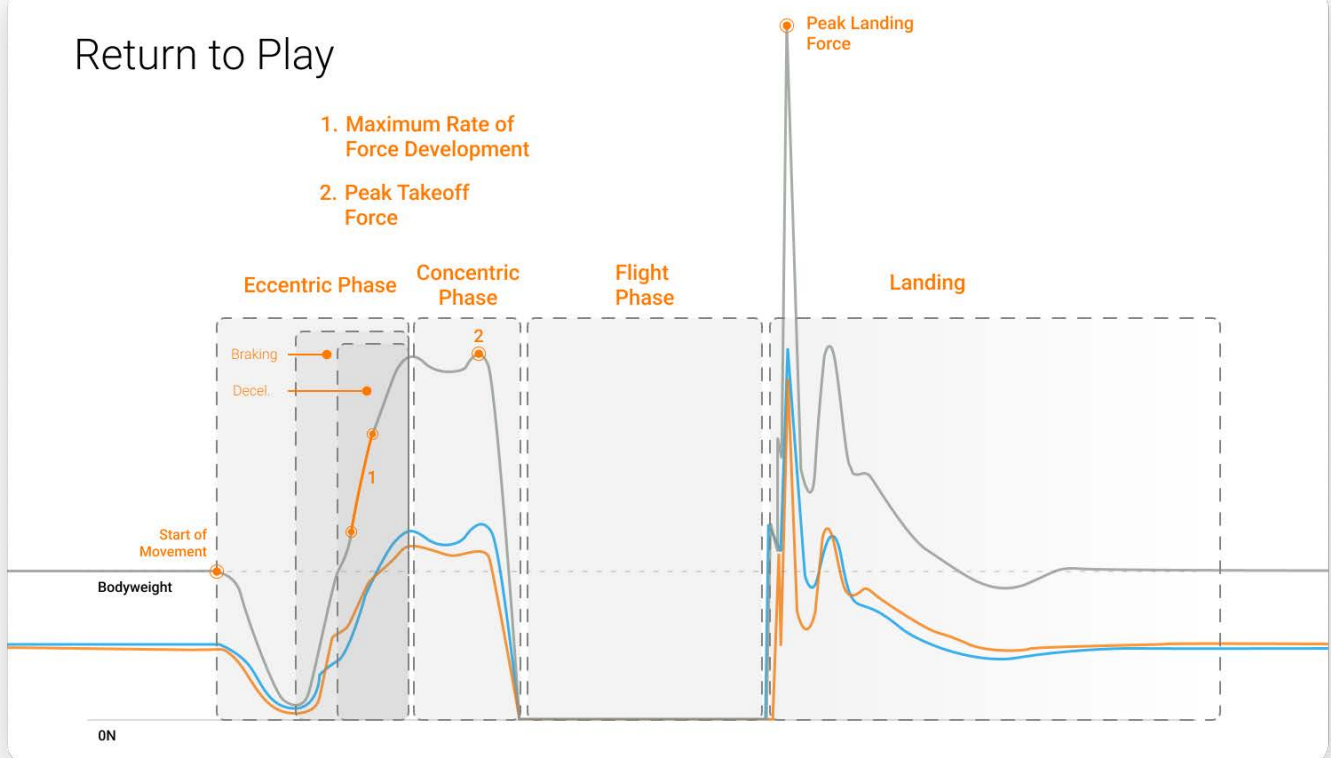
In the gym, the focus shifted to full force-velocity curve exposure, with an emphasis on increasing neurocognitive challenges. This includes incorporating variability through task constraints

and perturbations, targeting skills such as attention, pattern recognition, anticipation, decision-making and visual disruption.

Training also integrates a range of cues – visual, auditory, tactile – and multi-planar, evasive tasks to further enhance situational exposure.

CMJ characteristics demonstrate improvements in eccentric deceleration symmetry, force at zero velocity and concentric performance, noting improved preparation to safely return to soccer performance.

Return to Play



Key Takeaways and Summary

Vertical jump testing and force plate analysis are essential for enhancing return-to-play decisions and identifying performance deficits to target during rehabilitation.

Monitoring these metrics regularly (when appropriate for tissue healing) helps ensure that stimuli drive the right adaptations and confirms the rehabilitation program is addressing key performance indicators (KPIs).

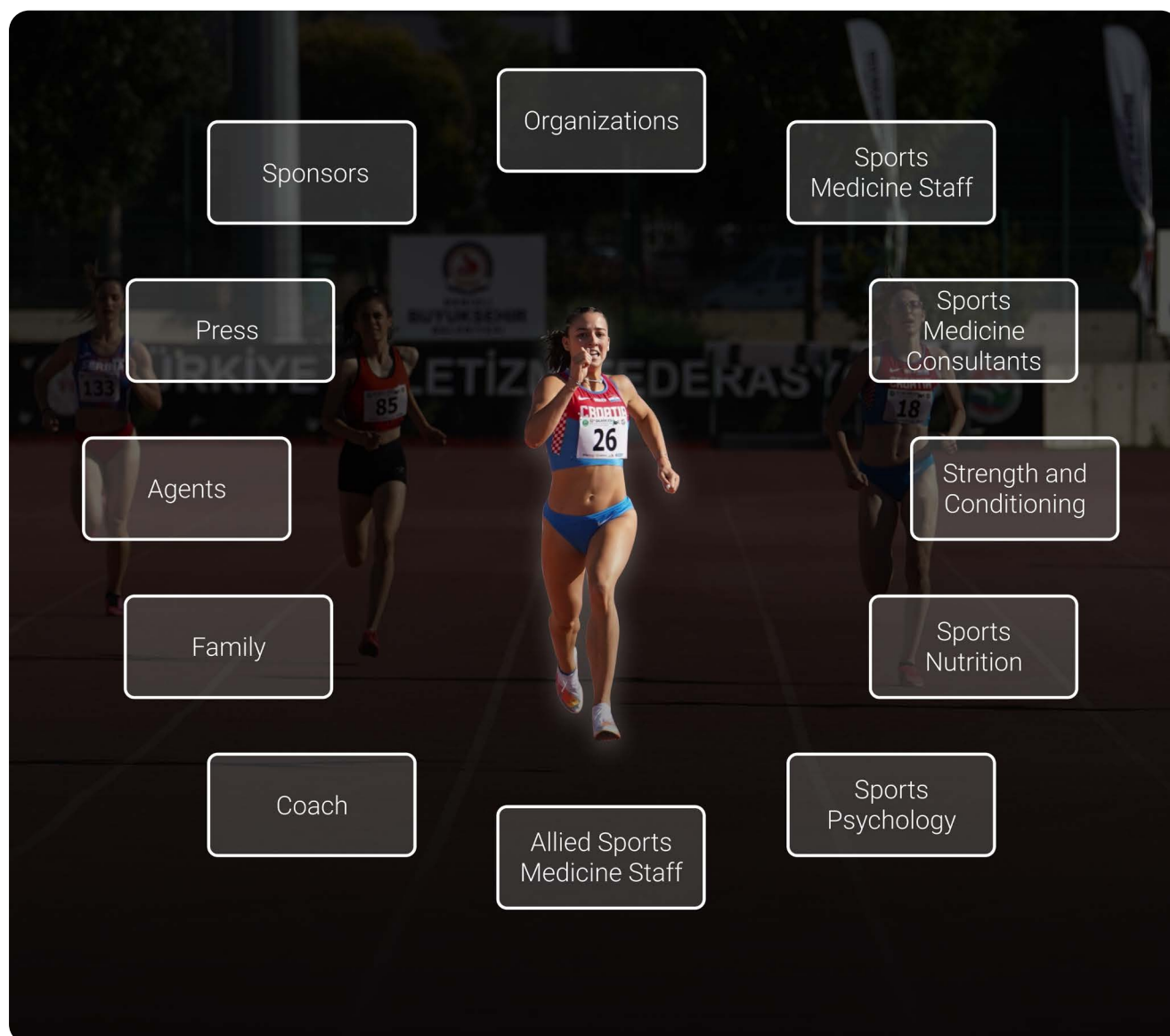
Focusing on valid, reliable outputs and understanding the drivers that underpin them allows practitioners to interpret force-time curves, identify deficiencies and make data-driven decisions to address performance gaps.

What Next?

Rehabilitation aims to return athletes to sport at their highest level with minimal reinjury risk. However, this is a continuous journey, not defined by isolated tests at a single time point.

Recovery after ACL injury involves:

- Developing physical capacity
- Learning or re-learning skills
- Progressing from controlled to chaotic environments
- Gradually increasing load to drive adaptation
- Building psychological readiness



The range of support staff involved in the long-term rehabilitation and return-to-sport process.

While this guide addresses select components, ongoing monitoring and adaptation are essential throughout rehabilitation. A multidisciplinary team with clear roles and effective communication is key, keeping the athlete at the center through shared decision-making.

To stay at the forefront of ACL care, practitioners must continuously update their skills.

Tips for **staying ahead**

- **Networking & Collaboration** – Engage with physiotherapists, surgeons and sports scientists for new insights.
- **Continuing Education** – Attend webinars and courses to enhance knowledge.
 - **Webinar:** [Technology in ACL Rehabilitation](#)
 - **Webinar:** [Force Plate Analysis Post-ACLR](#)

Networks Worth Following

ORKA Performance



Mick Hughes



Athletes Authority



Jane Rooney



Aspetar



Dustin Grooms



Roula Kotsifaki



Tim McGrath



Enda King



Erik Meira



Learn.Physio



Paul Read



Amy Arundale



Nicole Surdyka

